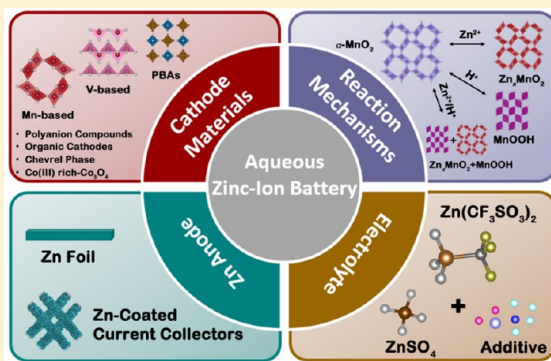


Recent Advances in Aqueous Zinc-Ion Batteries

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ABSTRACT: Although current high-energy-density lithium-ion batteries (LIBs) have taken over the commercial rechargeable battery market, increasing concerns about limited lithium resources, high cost, and insecurity of organic electrolyte scale-up limit their further development. Rechargeable aqueous zinc-ion batteries (ZIBs), an alternative battery chemistry, have paved the way not only for realizing environmentally benign and safe energy storage devices but also for reducing the manufacturing costs of next-generation batteries. This Review underscores recent advances in aqueous ZIBs; these include the design of a highly reversible Zn anode, optimization of the electrolyte, and a wide range of cathode materials and their energy storage mechanisms. We also present recent advanced techniques that aim at overcoming the current issues in aqueous ZIB systems. This Review on the future perspectives and research directions will provide a guide for future aqueous ZIB study.



Electrochemical energy storage devices, such as rechargeable batteries, are critical for overcoming the worldwide energy challenge.¹ Because of their high energy density and long-cycle-life, nonaqueous lithium-ion batteries (LIBs) have dominated in portable electronics and emerging electric/hybrid vehicles and now are considered as the possible choice for future electric vehicles and grid-scale energy storage systems.² However, the increasing concerns about limited lithium resources, high-cost, and safety issue strongly limit their further development for large-scale applications.³ Sodium-ion batteries (SIBs) and potassium-ion batteries (KIBs) are plausible alternatives to LIBs because these devices are based on relatively abundant and cheap sodium (potassium) elements and their similar chemical properties to lithium, but they suffer from the low energy density, use highly toxic and flammable electrolytes, and have high operating cost and security issues.^{4–6} The drawbacks of these organic-based systems motivate us to explore alternative battery chemistry with low-cost, high safety, and long cycle life.

Aqueous rechargeable batteries are a promising class of batteries for grid-scale electrochemical energy storage because of their low-cost, high operational safety, and environmental benignity.⁷ Furthermore, the aqueous electrolytes offer 2 orders of magnitude higher ionic conductivities ($\sim 1 \text{ S cm}^{-1}$) than that of nonaqueous electrolytes ($\sim 1\text{--}10 \text{ mS cm}^{-1}$).⁸ To date, a variety of aqueous batteries based on naturally abundant alkali metal cations (e.g., Na^+ and K^+)^{9,10} and multivalent charge carriers (e.g., Mg^{2+} , Al^{3+} , and Zn^{2+}) are under development.¹¹

Obviously, the systems employing multivalent ions can, in principle, achieve higher specific capacity and energy density because of the multiple electrons involved in redox reactions.^{12,13} The development of aqueous magnesium-ion batteries (MIBs) is restricted because the available cathode materials for this system have been limited to very few compounds owing to the sluggish Mg^{2+} diffusion in host lattices.¹⁴ In addition, the passivation of the Mg anode greatly prevents further transport of Mg^{2+} ions, requiring a usable anode material to build more powerful aqueous MIBs.^{11,15,16} Aqueous aluminum-ion batteries (AIBs) have also been proposed because of the high volumetric capacity (ca. $8040 \text{ mA h cm}^{-3}$) and good gravimetric capacity (ca. 2980 mA h g^{-1}) of the aluminum anode.^{17–19} Nonetheless, because of the formation of a protective oxide (Al_2O_3) film on the anode in aqueous electrolytes ($4 < \text{pH} < 8$) accompanied by a decrease in cell efficiency and electrode potential, as well as the tendency of aluminum to corrode unevenly,^{19–21} aqueous AIBs are still in the experimental stage and are limited to the development of primary high-energy systems.

In this regard, aqueous zinc-ion batteries (ZIBs) with mild neutral pH (or slightly acidic) electrolyte hold particular promise for grid-scale energy storage.²² Since Volta et al. employed metallic zinc (Zn) in the first battery in 1799, Zn anode has been

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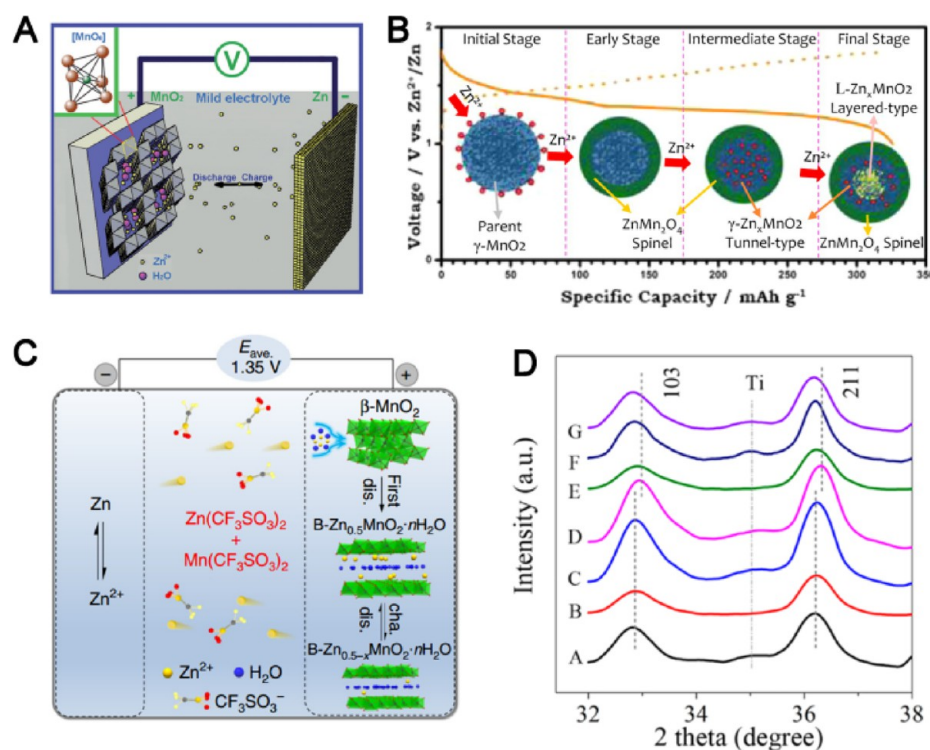


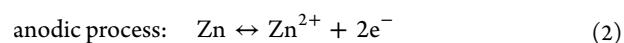
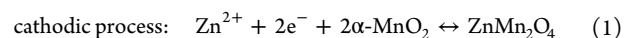
Figure 1. Typical Examples of manganese-based electrode materials based on Zn^{2+} insertion/extraction mechanism. (A) Schematics of the chemistry of the aqueous ZIBs, in which the migration of Zn^{2+} ions are between tunnels of $\alpha\text{-MnO}_2$ cathode and Zn anode. Reprinted with permission from ref 22. Copyright 2012 Wiley-VCH. (B) Schematic illustration of Zn^{2+} insertion in $\gamma\text{-MnO}_2$ cathode. Reprinted from ref 49. Copyright 2015 American Chemical Society. (C) Schematic illustration of reactions in aqueous Zn/ $\beta\text{-MnO}_2$ cell. Reprinted with permission from ref 56. Copyright 2017 Macmillan Publishers. (D) Ex situ XRD patterns of ZnMn_2O_4 cathode. Reprinted from ref 33. Copyright 2016 American Chemical Society.

regarded as an ideal negative electrode in various primary and secondary Zn-based batteries, such as Zn–Mn, Ni–Zn, and Zn–air batteries, because of its high theoretic capacity, relatively low redox potential, low cost, and high safety.^{23–25} Among them, alkaline Zn– MnO_2 batteries show great potential and have become dominant in primary battery chemistry.^{8,26} Later efforts developed rechargeable alkaline Zn– MnO_2 batteries; unfortunately, they were plagued by poor cycle life and inferior discharge performance due to the formation of zinc dendrite and irreversible discharged species.^{27–29} In 1988, Shoji et al. first reported a rechargeable aqueous Zn– MnO_2 battery using mild neutral or slightly acidic electrolyte (ZnSO_4 aqueous electrolyte),³⁰ which opened the door to aqueous zinc-ion batteries.

To date, many significant works regarding aqueous ZIBs have been reported in the literature, including metallic zinc (Zn) anode^{31,32} and cathode materials (e.g., manganese-based oxides,^{22,33–35} Prussian blue analogs,^{36,37} vanadium-based oxides,^{38–40} polyanion compounds,^{41,42} Chevrel phase compounds,^{43,44} sustainable quinone analogs,^{45,46} etc.). Recently, Fan et al. summarized part of the cathode material and their reaction mechanisms as well as their electrolytes.⁴⁷ In this Review, the energy storage mechanisms in these systems are systematically analyzed and summarized. Furthermore, the merits and issues as well as the optimization strategies related to various cathodes, Zn anode, and aqueous electrolytes are discussed. Finally, some potential future research directions from our personal perspectives or other related research topics are also presented. This Review provides a comprehensive overview focusing on the recent progress, challenges, and future perspectives in aqueous ZIBs.

Energy Storage Mechanisms in Aqueous ZIB Systems. Unlike the well-established lithium/sodium-ion-based energy storage chemistries (e.g., insertion–conversion–alloying reaction mechanism) for storage monovalent alkali metal cations,⁴⁸ the reaction mechanisms in aqueous ZIB systems are complicated and controversial. To date, the reaction mechanisms involved remain a topic of discussion and are underdeveloped. According to the literature, the redox reactions in aqueous ZIB systems mainly involve three mechanisms: Zn^{2+} insertion/extraction,^{22,38,49} chemical conversion reaction,⁵⁰ and $\text{H}^+/\text{Zn}^{2+}$ insertion/extraction,^{51,52} which are summarized and discussed in this section.

Zn^{2+} Insertion/Extraction Mechanism. Many compounds with tunnel-type and layered-type structure enable the insertion/extraction of Zn^{2+} ions into/from their hosts owing to the small ionic radii of Zn^{2+} (0.74 Å). For example, Kang et al. reported an aqueous ZIB, which is composed of an $\alpha\text{-MnO}_2$ cathode, a zinc anode, and a mild ZnSO_4 or $\text{Zn}(\text{NO}_3)_2$ aqueous electrolyte (Figure 1A),²² demonstrating that the charge storage mechanism in the aqueous Zn/ $\alpha\text{-MnO}_2$ system is based on the migration of Zn^{2+} ions between tunnels of $\alpha\text{-MnO}_2$ cathode and Zn anode, namely a typical Zn^{2+} insertion/extraction reaction mechanism as follows:



Soon afterward, Kang's group further revealed Zn^{2+} insertion into the host of α -, β -, γ -, and δ -type MnO_2 with different tunnel structures as their main charge storage mechanism.⁵³

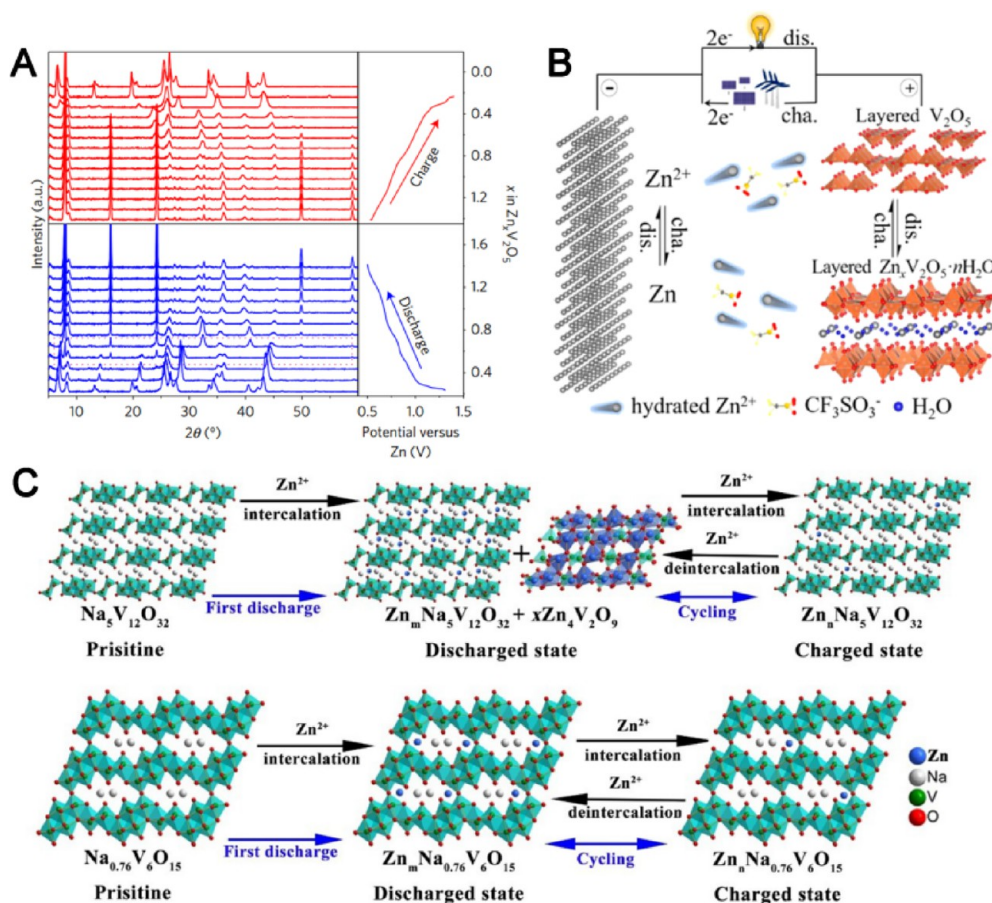
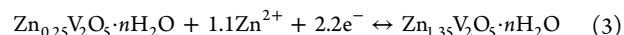


Figure 2. Typical examples of vanadium-based electrode materials based on Zn^{2+} insertion/extraction mechanism. (A) Operando XRD patterns. Reprinted with permission from ref 38. Copyright 2016 Macmillan Publishers. (B) Schematic illustration of the rechargeable aqueous $\text{Zn}-\text{V}_2\text{O}_5$ battery chemistry. Reprinted from ref 58. Copyright 2018 American Chemical Society. (C) Schematic illustration of Zn^{2+} insertion/extraction process of $\text{Na}_5\text{V}_{12}\text{O}_{32}$ and $\text{Na}_{0.76}\text{V}_6\text{O}_{15}$ during cycling. Reprinted with permission from ref 64. Copyright 2018 Wiley-VCH.

It should be noted that some studies also focused on Zn^{2+} insertion into the $\alpha\text{-MnO}_2$ host, but they demonstrated a phase transition from tunneled structure ($\alpha\text{-MnO}_2$) to layered Zn -buserite or birnessite,^{54,55} not the spinel ZnMn_2O_4 phase mentioned above. Kim et al. reported an aqueous $\text{Zn}/\gamma\text{-MnO}_2$ cell in which $\gamma\text{-MnO}_2$ undergoes a complex multiphase transformation on successive insertion of Zn^{2+} .⁴⁹ As shown in Figure 1B, in the early stages, partial $\gamma\text{-MnO}_2$ is transferred into spinel-type ZnMn_2O_4 . The further inserted Zn^{2+} ions occupy the 1×2 tunnels of $\gamma\text{-MnO}_2$ to form a tunnel-type $\gamma\text{-Zn}_x\text{MnO}_2$ phase. As more Zn^{2+} ions are inserted, Zn -containing tunnels in a portion of $\gamma\text{-Zn}_x\text{MnO}_2$ tend to expand and open up the structural framework to form a layered-type $\text{L-Zn}_y\text{MnO}_2$ phase. For the $\beta\text{-MnO}_2$, it seems to hardly incorporate Zn^{2+} ions because of its narrow 1×1 tunnels;⁵³ however, Chen et al. elucidated that a layer-type $\text{B-Zn}_x\text{MnO}_2 \cdot n\text{H}_2\text{O}$ phase was generated during the initial discharge, followed by reversible insertion/extraction of Zn^{2+} ions in this layered structure (Figure 1C).⁵⁶ In some of them, the insertion/extraction of Zn^{2+} is homogeneous (e.g., α - and $\beta\text{-MnO}_2$), while in others, it is heterogeneous (e.g., $\gamma\text{-MnO}_2$). The different inserted behaviors of Zn^{2+} ions in these MnO_2 may be attributed to the difference in crystallographic polymorphs that change the ion insertion thermodynamics and kinetics.⁵⁷ In addition to MnO_2 , other manganese-based oxides have also been investigated. For instance, Chen et al. reported a cation-deficient spinel ZnMn_2O_4 cathode with

highly reversible insertion/extraction of Zn^{2+} .³³ The ex situ XRD patterns were conducted at different charge/discharge states to elucidate the evolution of ZnMn_2O_4 cathode. As shown in Figure 1D, the diffraction peaks gradually shift to higher angle when charged from 0.8 to 1.9 V, demonstrating the shrink of lattice spacing after the extraction of Zn^{2+} . As expected, the diffraction peaks are recovered upon full discharging back to 0.8 V, suggesting good reversibility of the ZnMn_2O_4 cathode.

In addition to the progress of manganese-based oxides, other materials with Zn^{2+} insertion reaction mechanism are also being intensively studied. Nazar et al. opened a new chapter in the $\text{Zn}/\text{vanadium-based oxides}$ battery, which is constructed by a $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ cathode with a Zn^{2+} (de)intercalation mechanism.³⁸ At the initial state, the structure of pristine $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ was modified via water intercalation upon immersion in the electrolyte, which results in an increase in the interlayer distance, thus expanding the galleries for facile Zn^{2+} intercalation. As the discharge/charge reaction proceeds, the insertion/extraction of 1.1 Zn^{2+} ions into/from $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ is highly reversible, corresponding to the following equation:



However, this process is complex as elucidated from an operando XRD investigation (Figure 2A). In detail, two solid-solution regimes ($0.25 < x < 0.55$ and $0.55 < x < 1.4$) are separated by a phase transition in which the structural water

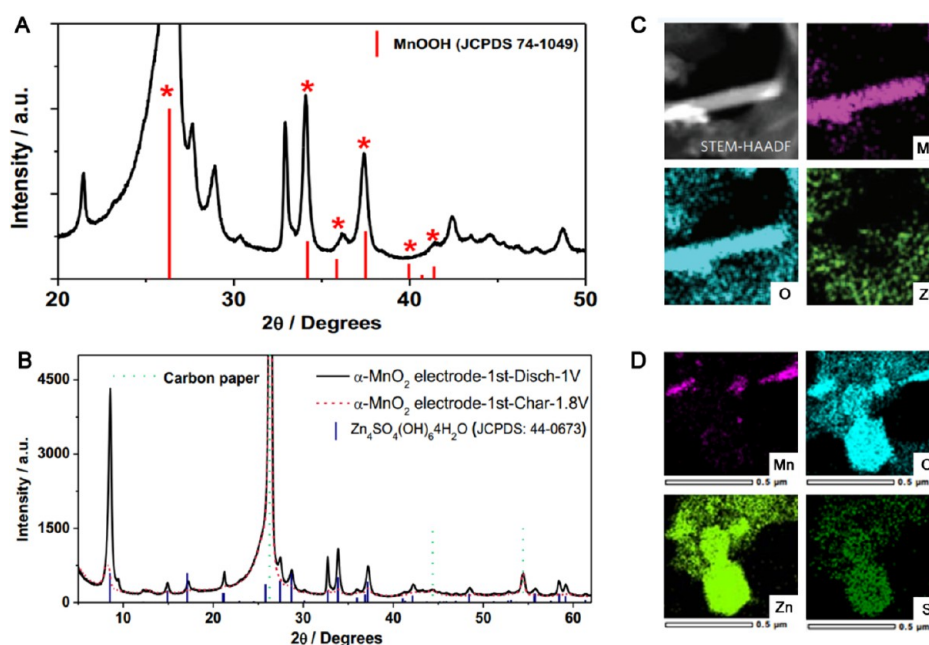


Figure 3. Typical examples of battery electrode materials based on chemical conversion reaction mechanism. (A) XRD pattern of α - MnO_2 electrode discharged to 1 V in first cycle, which verifies the formation of MnOOH phase. (B) XRD patterns of α - MnO_2 electrodes discharged to 1 V and charged back to 1.8 V in first cycle, which verify the formation of $\text{ZnSO}_4[\text{Zn}(\text{OH})_2]_3 \cdot x\text{H}_2\text{O}$ phase. STEM-EDS mappings of the elemental distributions of (C) Mn, O, and Zn in the MnO_2 electrode in the initial fully discharged state and (D) Mn, O, Zn, and S in the MnO_2 electrode in the initial fully charged state. Reprinted with permission from ref 50. Copyright 2016 Macmillan Publishers.

in $\text{Zn}_x\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ exchanged with Zn^{2+} ions (insertion of $\sim 0.3 \text{ Zn}^{2+}$). A small contraction of the interlayer distance is observed at the prior solid-solution regime (namely, insertion of $< 0.3 \text{ Zn}^{2+}$), which is due to the increased interlayer electrostatic interaction as the Zn^{2+} content increases. Then, the new set of (00 l) reflections show an increase in their intensities, accomplished by the intensity and position evolution of the other peaks, strongly indicating that the structure continues to evolve upon Zn^{2+} insertion. The charge process follows the reverse evolution of the discharge process. Since then, a series of vanadium-based materials, such as $\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$,^{39,58} $\text{Zn}_3\text{V}_2\text{O}_7(\text{OH})_2 \cdot 2\text{H}_2\text{O}$,⁴⁰ $\text{Ca}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$,⁵⁹ $\text{Na}_2\text{V}_6\text{O}_{16} \cdot 3\text{H}_2\text{O}$,^{60,61} $\text{Na}_{0.33}\text{V}_2\text{O}_5$,⁶² VS_2 ,⁶³ etc., based on Zn^{2+} intercalation chemistry have been investigated. For example, Zhang et al. reported a layered V_2O_5 cathode, which allows a high reversibility for hydrated Zn^{2+} (de)intercalation (Figure 2B).⁵⁸ It demonstrates that the coinsertion of Zn^{2+} and H_2O results in a structural evolution from layered V_2O_5 to layered $\text{Zn}_x\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$, but a reverse structural evolution was observed during the extraction of hydrated Zn^{2+} , indicating good structural reversibility. Very recently, Liang et al. demonstrated that different structures and configurations play a key role in the storage/release behaviors of Zn^{2+} ions.^{64,65} For instance, they constructed two typical sodium vanadates with NaV_3O_8 -type layered structure, e.g., $\text{Na}_5\text{V}_{12}\text{O}_{32}$ ($\text{Na}_{1.25}\text{V}_3\text{O}_8$), $\text{HNaV}_6\text{O}_{16} \cdot 4\text{H}_2\text{O}$ ($\text{H}_{0.5}\text{Na}_{0.5}\text{V}_3\text{O}_8 \cdot 2\text{H}_2\text{O}$), and β - $\text{Na}_{0.33}\text{V}_2\text{O}_5$ -type tunneled structure, e.g., $\text{Na}_{0.76}\text{V}_6\text{O}_{15}$ ($\text{Na}_{0.253}\text{V}_2\text{O}_5$), indicating that the structural destruction and the formation of the second phase were observed for the layered $\text{Na}_5\text{V}_{12}\text{O}_{32}$ and $\text{HNaV}_6\text{O}_{16} \cdot 4\text{H}_2\text{O}$, while the tunneled $\text{Na}_{0.76}\text{V}_6\text{O}_{15}$ exhibits reversible insertion/extraction of Zn^{2+} without structural collapse (Figure 2C). It is reported that layered NaV_3O_8 possesses single-connected and triconnected oxygen atoms on the layer surface, while the tunneled β - $\text{Na}_{0.33}\text{V}_2\text{O}_5$ contains only single-connected oxygen atoms that can provide stronger interaction with the Na^+ than

that of triconnected oxygen atoms.^{66,67} As result, some part of Na^+ in $\text{Na}_5\text{V}_{12}\text{O}_{32}$, which bonds with triconnected oxygen atoms, is more likely to be displaced by the Zn^{2+} to form a new phase of $\text{Zn}_4\text{V}_2\text{O}_9$. In contrast, the Na^+ in $\text{Na}_{0.76}\text{V}_6\text{O}_{15}$ bonding with single-connected oxygen atoms acts as strong “pillars” to stabilize the whole structure and make it more difficult to be destroyed.⁶⁴

Because of their open framework, Prussian blue analogues (PBAs)^{36,37,68–71} and polyanionic olivine-based phosphates MNPO_4 ($\text{M} = \text{Li}, \text{Na}, \text{K}$; $\text{N} = \text{V}, \text{Fe}, \text{Mn}, \text{Ni}, \text{Co}$, etc.)^{41,72–74} have been also verified to enable the insertion/extraction of divalent Zn^{2+} ions. The cathode electrode reactions in these compounds involve the insertion/extraction of Zn^{2+} into/from their three-dimensional pathways, and anode electrode reactions are related to the deposition and dissolution of zinc.

Chemical Conversion Reaction Mechanism. Unlike most cases in which the capacity is induced by the reversible Zn^{2+} insertion/extraction, Liu and co-workers demonstrated a $\text{Zn}/\alpha\text{-MnO}_2$ system, which operates on the principle of a chemical conversion reaction mechanism between $\alpha\text{-MnO}_2$ and MnOOH .⁵⁰ As shown in Figure 3A, the MnOOH phase is observed at the fully discharged state, which is due to the reaction between $\alpha\text{-MnO}_2$ with a proton from water. To reach a neutral charge in the system, the sequent OH^- ions will react with ZnSO_4 and H_2O in the aqueous electrolyte to form a $\text{ZnSO}_4[\text{Zn}(\text{OH})_2]_3 \cdot x\text{H}_2\text{O}$ phase (Figure 3B). This phenomenon is strongly supported by the STEM-EDS mapping of the $\alpha\text{-MnO}_2$ electrode in the discharged state, which reveals that the short nanorods and nanoparticles consist of O and Mn (Figure 3C), while Zn is mainly distributed on the flakelike compounds (Figure 3D). The reaction mechanism can be formulated as below:

Cathodic process



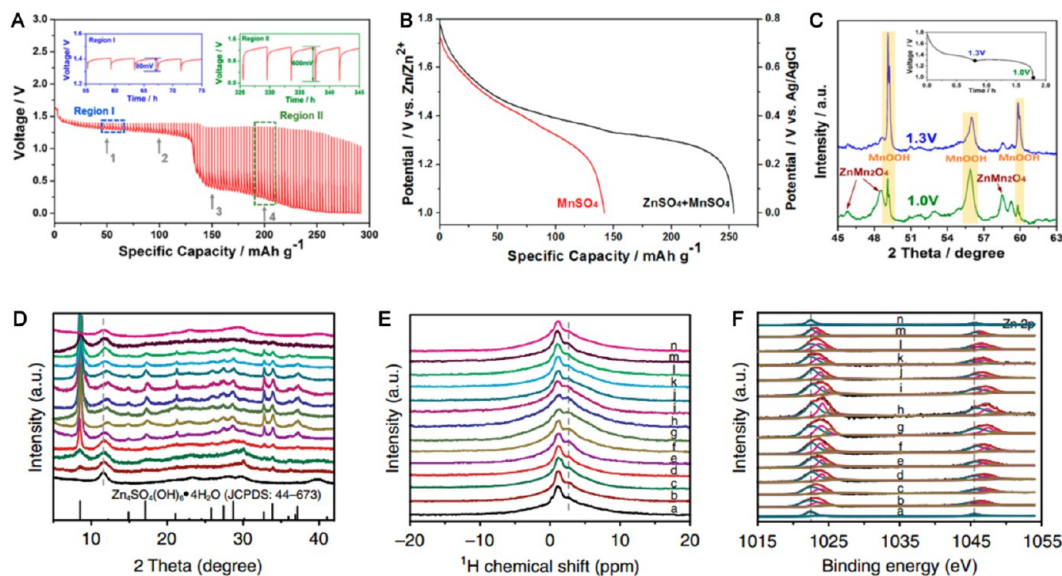
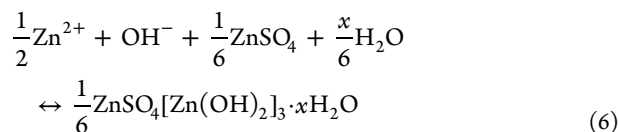


Figure 4. Typical examples of battery electrode materials based on H^+ and Zn^{2+} insertion/extraction mechanism. (A) Discharge galvanostatic intermittent titration technique (GITT) profiles of MnO_2 electrode, indicating two different reaction regions. (B) Discharge curves of MnO_2 electrode in 0.2 M MnSO_4 solution with or without ZnSO_4 as electrolytes, confirming that the main reaction in region II is contributed by the Zn^{2+} insertion. (C) Ex situ XRD patterns of the MnO_2 electrode cathode at depth of discharge at 1.3 and 1.0 V, respectively. Reprinted from ref 51. Copyright 2017 American Chemical Society. (D) Ex situ XRD patterns, (E) ex situ solid state ^1H NMR, and (F) XPS spectra of Zn 2p of $\text{NaV}_3\text{O}_8 \cdot 1.5\text{H}_2\text{O}$ electrodes. Reprinted with permission from ref 52. Copyright 2018 Macmillan Publishers.



Anodic process



As discussed above, several different mechanisms are observed for $\alpha\text{-MnO}_2$ in different literature contributions, suggesting more precise experiments and analysis methods are required to further clarify the reaction mechanism of $\alpha\text{-MnO}_2$.

H^+ and Zn^{2+} Insertion/Extraction Mechanism. The feasible host materials with open tunnel or layered framework may enable the coininsertion of H^+ and Zn^{2+} . Because of the different insertion thermodynamics and kinetics of H^+ and Zn^{2+} , Wang et al. proposed a consequent H^+ and Zn^{2+} insertion reaction mechanism for aqueous $\text{Zn}/\alpha\text{-MnO}_2$ battery.⁵¹ These two reactions happened in two regions, which are identified by the electrochemical and structural analysis. Discharge galvanostatic intermittent titration technique (GITT) profiles of the MnO_2 electrode (Figure 4A) show that the total overvoltage in region II is 0.6 V, which is much larger than that in region I (0.08 V). The significant difference in regions I and II should be attributed to different ion insertion. It is known that the voltage jump observed instantly after applying current is attributed to the ohm and charge transfer resistances, while the gradual voltage change after the discharge is mainly ascribed to the ion diffusion. The large overvoltage in region II is attributed to both large voltage jump and slow ion diffusion. Given the large size of Zn^{2+} and the fact that bivalent Zn^{2+} would exhibit strong electrostatic interactions with the host lattice, it is highly possible that the voltage plateau in region II

is mainly attributed to the slow Zn^{2+} insertion. This hypothesis is confirmed by the electrochemical behaviors of MnO_2 electrode in 0.2 M MnSO_4 electrolytes with or without ZnSO_4 . As shown in Figure 4B, the discharged plateau at 1.3 V in region II disappeared because of the absence of Zn^{2+} . A typical MnOOH phase is observed after discharge to 1.3 V because of the H^+ insertion, while the ZnMn_2O_4 phase can be observed only after further discharging to 1.0 V (Figure 4C), which strongly proved that the main reaction in region II is contributed by the Zn^{2+} insertion.

The coininsertion of H^+ and Zn^{2+} mechanism is also observed in the vanadium-based oxides ($\text{NaV}_3\text{O}_8 \cdot 1.5\text{H}_2\text{O}$, NVO) as proposed by Chen and co-workers.⁵² As depicted by ex situ XRD patterns (Figure 4D), the $\text{Zn}_4\text{SO}_4(\text{OH})_6 \cdot 4\text{H}_2\text{O}$ phase (JCPDS: 39-688) is successively formed during the discharge process, which is due to OH^- reacting with ZnSO_4 and H_2O in the aqueous electrolyte. To reach a neutral charge system, the extra H^+ would move into the NVO host to balance the electron-rich condition. The ex situ solid-state ^1H NMR analyses were conducted to confirm the continuous insertion of H^+ in the NVO during the discharge process (Figure 4E). Compared with the initial state, an extra peak at 2.7 ppm at the selected charge/discharge states would be ascribed to the H^+ that inserted in the NVO. In addition to the reversible insertion of H^+ , the Zn 2p spectra of NVO electrodes (Figure 4F) exhibit three pairs of Zn^{2+} peaks located at 1023/1046, 1024/1047, and 1025/1048 eV, which are assigned to the inserted Zn^{2+} in NVO, as well as the Zn^{2+} in $\text{Zn}(\text{OH})_2$ and ZnSO_4 from $\text{Zn}_4\text{SO}_4(\text{OH})_6 \cdot 4\text{H}_2\text{O}$, respectively,⁵² indicating that Zn^{2+} takes part in the energy storage in the $\text{Zn}/\text{NaV}_3\text{O}_8 \cdot 1.5\text{H}_2\text{O}$ system.

Different Types and Challenges of Cathode Materials. Challenges still exist in aqueous ZIBs to achieve great breakthroughs in practical applications. One of the major challenges in the development and application of aqueous ZIBs is the lack of suitable cathodes that can offer high capacity and good structure stability during the process of guest ion storage.

As a result, most of the literature contributions on the study of aqueous ZIBs are focused on exploring and developing cathode materials. At present, these cathode materials mainly include (1) manganese-based oxides, (2) vanadium-based oxides, (3) Prussian blue analogues, (4) olivine-based phosphates, (5) Chevrel phase compounds, and (6) sustainable quinone cathodes. Others like VS_2 ,⁶³ Co(III) rich- Co_3O_4 ,⁷⁵ etc. have also been explored as high-performance aqueous ZIB cathodes. The features of these species, such as energy storage mechanism, testing voltage, operating voltage, specific discharge capacity, and electrolyte used in aqueous ZIB systems, are summarized in Table 1. The relationship between operating voltage and specific capacity of these cathodes is shown visually in Figure 5. The merits and challenges of these cathode materials are also discussed in this section.

Manganese-Based Oxides. Manganese (Mn)-based oxides, which possess unique advantages, such as low-cost, abundant reserves, environmental friendliness, low toxicity, and multiple valence states (Mn^0 , Mn^{2+} , Mn^{3+} , Mn^{4+} and Mn^{7+}), have been considered as attractive energy storage materials.^{76–79} Recently, manganese-based oxides including MnO_2 with different crystal structures,^{22,50,51,53,80–82} Mn_2O_3 ,³⁵ Mn_3O_4 ,^{83,84} ZnMn_2O_4 ,³³ etc., are one of the most promising cathodes for aqueous ZIBs.

In the early state of aqueous ZIBs, extensive research efforts have been focused on MnO_2 for cathode materials because their tunnel or layered structure allows reversible insertion/extraction of Zn^{2+} ions. The family of MnO_2 materials have a variety of crystallographic polymorphs (e.g., α -, β -, γ -, δ -, ϵ -, and λ -type MnO_2), which are dependent on the type of linkage between the fundamental octahedral unit $[\text{MnO}_6]$.⁸⁵ The basic structural unit of α - MnO_2 is composed of $[\text{MnO}_6]$ octahedral double chains that are corner-sharing with adjacent double chains in the plane and form $(2 \times 2) + (1 \times 1)$ tunnel structures (Figure 6A), which can accommodate different cations or coordination ions.⁸⁶ The previous reported α - MnO_2 cathodes displayed the similar working voltages around 1.2–1.4 V with a capacity of more than 200 mA h g^{-1} . Among all the MnO_2 polymorphs, β - MnO_2 structure is regarded as the most thermodynamically stable structure, which is composed of $[\text{MnO}_6]$ octahedral single chains by vertice-sharing, forming (1×1) tunnel along c axis (Figure 6B). This narrow tunnel is usually not conducive to the diffusion of cations e.g. Li^+ and Zn^{2+} .⁵³ Recently, however, Chen et al. found that β - MnO_2 enabled a reversible insertion/extraction of Zn^{2+} ions through a phase transition to form a layer-type phase Zn-buserite during the initial discharge.⁵⁶ Remarkably, the cathode exhibits a high reversible capacity of 225 mA h g^{-1} and long-term cyclability with 94% capacity retention over 2000 cycles. According to Kim, the γ - MnO_2 phase with an irregular (1×2) tunnel (Figure 6C) can deliver a high capacity of 285 mA h g^{-1} at 0.05 mA cm^{-2} with a defined plateau at around 1.25 V vs Zn/Zn^{2+} .⁴⁹ As a typical layered compound, δ - MnO_2 (Figure 6D) is also investigated to be a potential cathode for aqueous ZIBs with a high capacity of 269 mA h g^{-1} at 100 mA g^{-1} using 1 mol L^{-1} ZnSO_4 as electrolyte.⁵³ In addition, the Zn^{2+} storage capability of ϵ - MnO_2 has been studied by Wang and co-workers.⁵¹ It is clear that the electrochemical behavior of MnO_2 strongly depends on its crystallographic forms.

However, as mentioned earlier, the most critical issue of MnO_2 is the energy storage mechanisms involved in Zn^{2+} insertion/extraction,^{22,54} chemical conversion reaction,⁵⁰ and $\text{H}^+/\text{Zn}^{2+}$ insertion/extraction,⁵¹ which are quite complicated

and need to be further clarified. Another major problem is the significant capacity fading of these MnO_2 materials due to dissolution of Mn^{2+} from Mn^{3+} disproportionation into the electrolyte during cycling.⁵⁰ As evidenced by Liu et al., a rapid deterioration in capacity was observed for the Zn/MnO_2 battery upon cycling in 2 M ZnSO_4 (Figure 6E).⁵⁰ Elemental analysis of dissolved Mn^{2+} ions shows that the Mn^{2+} ions gradually dissolved in the electrolyte but slowed after the initial 10 cycles. This is probably because the Mn^{2+} dissolution from the MnO_2 electrode increases Mn^{2+} concentration in the electrolyte and thus suppresses significant continuous Mn^{2+} dissolution. Therefore, preaddition of a certain amount of Mn^{2+} ions will provide an appropriate equilibrium between Mn^{2+} dissolution and the reoxidation of Mn^{2+} in the electrolyte, leading to high stability of the electrode. This method is confirmed to work in many cases, such as ZnSO_4 -based electrolyte (Figure 6F)⁵⁰ and $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ -based electrolyte (Figure 6G).⁵⁶

Except for the MnO_2 , other manganese-based oxides have also shown good Zn storage capability as cathode materials. For instance, Chen et al. reported a cation-deficient spinel ZnMn_2O_4 cathode for aqueous ZIBs using $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ electrolyte.³³ It is reported that the ideal spinel structure is unfit for Zn^{2+} insertion because of the high electrostatic repulsion among Zn^{2+} cations within the lattice.⁸⁷ Introducing abundant Mn vacancies in spinel ZnMn_2O_4 is verified to be a peculiar merit for Zn^{2+} diffusion in the spinel framework.³³ In a perfect spinel without Mn vacancies, Zn^{2+} ions migrate from one tetrahedral site (4a) to another by passing through an unoccupied octahedral site (8c), which is strongly impeded by a large electrostatic repulsion from Mn cations in a neighboring octahedral site (8d) (Figure 7A). However, the spinel rich in Mn vacancies allows for easier diffusion of Zn^{2+} without much electrostatic barrier, leading to higher diffusion coefficient of Zn^{2+} and consequently faster electrode kinetics (Figure 7B). In addition, Kang et al. enriched the manganese-based cathode family material by exploring two manganese oxides (e.g., α - Mn_2O_3 ³⁵ and Mn_3O_4 ⁸³). Following work by Liang et al. demonstrated that combining Mn_3O_4 with three-dimensional (3D) conductive substrates can improve its rate capability and long-term cycling performance (Figure 7C).⁸⁴ Very recently, Kim et al. developed a new manganese-based cathode MgMn_2O_4 with high voltage and high energy (Figure 7D).⁸⁸ The results indicate that partial Mg^{2+} ions escape from the spinel during the initial charge process, while the stable intercalation of $\text{Zn}^{2+}/\text{Mg}^{2+}$ into the spinel was observed in the subsequent cycles, strongly supporting the conclusion that the electrochemical extraction/insertion of cations into the spinel is reversible.

Prussian Blue Analogs. Prussian blue analogues (PBAs) exhibit an open-framework structure, in which the $\text{Fe}(\text{III})\text{C}_6$ octahedra are linked to $\text{M}(\text{II})\text{N}_4$ ($\text{M} = \text{Zn}^{2+}$, Ni^{2+} , Cu^{2+} , etc.) tetrahedra via CN ligands to form a porous 3D framework with large open A sites (Figure 8A).⁸⁹ Such a framework endows PBAs with the possibility of behaving as intercalation hosts for various cations. Many studies focus on the PBA family as hosts for monovalent or divalent ion batteries in both aqueous and organic solutions.^{89–92}

Recently, Zhang et al. reported a zinc hexacyanoferrate ($\text{Zn}_3[\text{Fe}(\text{CN})_6]_2$, ZnHCF) as cathode in aqueous ZIB systems using ZnSO_4 electrolyte.³⁶ ZnHCF yields a higher Zn^{2+} -intercalation voltage leading to an average operation voltage of ca. 1.7 V when combined with a zinc anode (Figure 8B).

Table 1. Summary of Features of Various Cathode Materials in Aqueous ZIBs, Such as Energy Storage Mechanisms, Testing Voltage, Working Voltage, Discharge Capacity, and Electrolyte Components

cathode material	reaction mechanism	testing voltage	main operating voltage (V)	discharge capacity	electrolyte components
α -MnO ₂ ⁵⁰	conversion reaction	1.0–1.8 V	1.35 and 1.25	285 mA h g ⁻¹ (C/3)	2 M ZnSO ₄ , 0.1 M MnSO ₄
α -MnO ₂ ⁵⁵	Zn ²⁺ insertion/extraction	0.7–2.0 V	1.20	210 mA h g ⁻¹ (21 mA g ⁻¹)	1 M ZnSO ₄
α -MnO ₂ ⁸¹	Zn ²⁺ insertion/extraction	0.7–2.0 V	1.40 and 1.30	255 mA h g ⁻¹ (50 mA g ⁻¹)	1 M ZnSO ₄
α -MnO ₂ ¹³⁵	not mentioned	1.0–2.0 V	1.30	302.1 mA h g ⁻¹ (60 mA g ⁻¹)	2 M ZnSO ₄ , 0.1 M MnSO ₄
α -MnO ₂ ⁵¹	H ⁺ /Zn ²⁺ insertion/extraction	1.0–1.8 V	1.40 and 1.30	290 mA h g ⁻¹ (90 mA g ⁻¹)	2 M ZnSO ₄ , 0.2 M MnSO ₄
α -MnO ₂ ²²	Zn ²⁺ insertion/extraction	1.0–1.9 V	1.30	210 mA h g ⁻¹ (0.5 C)	Zn(NO ₃) ₂
α -MnO ₂ /graphene ⁸⁰	Zn ²⁺ insertion/extraction	1.0–1.85 V	1.25 and 1.38	362.2 mA h g ⁻¹ (300 mA g ⁻¹)	2 M ZnSO ₄ , 0.2 M MnSO ₄
β -MnO ₂ ⁵⁶	Zn ²⁺ insertion/extraction	0.8–1.9 V	1.35 and 1.17	307 mA h g ⁻¹ (0.32 C)	3 M Zn(CF ₃ SO ₃) ₂
δ -MnO ₂ ⁵³	Zn ²⁺ insertion/extraction	1.0–2.0 V	1.30	269 mA h g ⁻¹ (100 mA g ⁻¹)	1 M ZnSO ₄
γ -MnO ₂ ⁴⁹	Zn ²⁺ insertion/extraction	1.0–1.8 V	1.30 and 1.20	285 mA h g ⁻¹ (0.05 mA cm ⁻²)	1 M ZnSO ₄
Mn ₂ O ₃ ³⁵	Zn ²⁺ insertion/extraction	1.0–1.9 V	1.34 and 1.23	150 mA h g ⁻¹ (100 mA g ⁻¹)	2 M ZnSO ₄
MnO ₂ ¹³³	not mentioned	1.0–1.8 V	1.37	366.6 mA h g ⁻¹ (74 mA g ⁻¹)	2 M ZnCl ₂ , 0.4 M MnSO ₄
Na _{0.95} MnO ₂ ⁸²	Na ⁺ insertion/extraction	1.0–2.0 V	1.40	60 mA h g ⁻¹ (2C)	0.5 M Zn(CH ₃ COO) ₂ , 0.5 M CH ₃ COONa
ZnMn ₂ O ₄ ³³	Zn ²⁺ insertion/extraction	0.8–2.0 V	1.35 and 1.10	150 mA h g ⁻¹ (50 mA g ⁻¹)	3 M Zn(CF ₃ SO ₃) ₂
MgMn ₂ O ₄ ⁸⁸	Zn ²⁺ /Mg ²⁺ insertion/extraction	0.5–1.9 V	1.36 and 1.16	130 mA h g ⁻¹ (100 mA g ⁻¹)	1 M (ZnSO ₄ + Mg SO ₄), 0.1 M MnSO ₄
Mn ₃ O ₄ ⁸⁴	conversion reaction	1.0–1.8 V	1.36 and 1.23	296 mA h g ⁻¹ (100 mA g ⁻¹)	2 M ZnSO ₄ , 0.1 M MnSO ₄
Mn ₃ O ₄ ⁸³	Zn ²⁺ insertion/extraction	0.8–1.9 V	1.31 and 1.18	239.2 mA h g ⁻¹ (100 mA g ⁻¹)	2 M ZnSO ₄
VO ₂ ¹⁰⁸	Zn ²⁺ insertion/extraction	0.3–1.5 V	0.90 and 0.52	357 mA h g ⁻¹ (100 mA g ⁻¹)	3 M Zn(CF ₃ SO ₃) ₂
V ₃ O ₇ ·H ₂ O ¹¹⁰	Zn ²⁺ insertion/extraction	0.4–1.1 V	0.80 and 0.60	400 mA h g ⁻¹ (375 mA g ⁻¹)	1 M ZnSO ₄ in H ₂ O, 0.25 M Zn(CF ₃ SO ₃) ₂
V ₂ O ₅ ¹⁰⁷	Li ⁺ insertion/extraction	0.2–1.6 V	1.10 and 0.90	242 mA h g ⁻¹ (50 mA g ⁻¹)	1 m Zn(CF ₃ SO ₃) ₂ and 21 m LiTFSI
V ₂ O ₅ ⁵⁸	Zn ²⁺ insertion/extraction	0.2–1.6 V	0.72	470 mA h g ⁻¹ (200 mA g ⁻¹)	3 M Zn(CF ₃ SO ₃) ₂
V ₂ O ₅ ¹⁰⁶	Zn ²⁺ insertion/extraction	0.4–1.4 V	0.7	372 mA h g ⁻¹ (0.3 A g ⁻¹)	3 M ZnSO ₄
V ₂ O ₅ ·nH ₂ O ³⁹	Zn ²⁺ insertion/extraction	0.2–1.6 V	0.91 and 0.54	381 mA h g ⁻¹ (60 mA g ⁻¹)	3 M Zn(CF ₃ SO ₃) ₂
H ₂ V ₃ O ₈ /graphene ¹¹¹	Zn ²⁺ insertion/extraction	0.2–1.6 V	0.80–0.60 and 0.60–0.40	336 mA h g ⁻¹ (100 mA g ⁻¹)	3 M Zn(CF ₃ SO ₃) ₂
Na _{0.33} V ₂ O ₅ ⁶²	Zn ²⁺ insertion/extraction	0.2–1.6 V	1.10 and 0.79, 0.63 and 0.52	373.2 mA h g ⁻¹ (0.2 A g ⁻¹)	3 M Zn(CH ₃ F ₃ SO ₃) ₂
Na ₂ V ₆ O ₁₆ ·1.63H ₂ O ⁶⁰	Zn ²⁺ insertion/extraction	0.2–1.6 V	0.64 and 0.46	352 mA h g ⁻¹ (50 mA g ⁻¹)	3 M Zn(CF ₃ SO ₃) ₂
Na ₂ V ₆ O ₁₆ ·3H ₂ O ⁶¹	Zn ²⁺ insertion/extraction	0.4–1.4 V	0.75 and 0.51	361 mA h g ⁻¹ (100 mA g ⁻¹)	1 M ZnSO ₄
NaV ₃ O ₈ ·1.5H ₂ O ⁵²	H ⁺ /Zn ²⁺ insertion/extraction	0.3–1.25 V	0.85 and 0.55	380 mA h g ⁻¹ (50 mA g ⁻¹)	1 M ZnSO ₄ or 1 M ZnSO ₄ /1 M Na ₂ SO ₄
Na ₅ V ₁₂ O ₃₂ ⁶⁴	Zn ²⁺ insertion/extraction	0.4–1.4 V	~0.75	281 mA h g ⁻¹ (50 mA g ⁻¹)	2 M ZnSO ₄
K ₂ V ₈ O ₂₁ ⁶⁵	Zn ²⁺ insertion/extraction	0.4–1.4 V	0.91 and 0.51	216 mA h g ⁻¹ (500 mA g ⁻¹)	2 M ZnSO ₄
Zn _{0.25} V ₂ O ₅ ·nH ₂ O ³⁸	Zn ²⁺ insertion/extraction	0.5–1.4 V	~0.9	300 mA h g ⁻¹ (50 mA g ⁻¹)	1 M ZnSO ₄
Zn ₃ V ₂ O ₇ (OH) ₂ ·2H ₂ O ⁴⁰	Zn ²⁺ insertion/extraction	0.2–1.8 V	1.28, 0.84, 0.58	213 mA h g ⁻¹ (50 mA g ⁻¹)	1 M ZnSO ₄
Zn ₂ V ₂ O ₇ ¹⁰²	Zn ²⁺ insertion/extraction	0.4–1.4 V	0.81 and 0.61	190.5 mA h g ⁻¹ (300 mA g ⁻¹)	1 M ZnSO ₄
Ca _{0.24} V ₂ O ₅ ·0.83H ₂ O ⁵⁹	Zn ²⁺ insertion/extraction	0.6–1.6 V	0.7 and 1.2	340 mA h g ⁻¹ (0.2 C)	1 M ZnSO ₄
Na ₃ V ₂ (PO ₄) ₃ ⁴¹	Zn ²⁺ insertion/extraction	0.8–1.7 V	1.15	97 mA h g ⁻¹ (0.5 C)	0.5 M Zn(CH ₃ COO) ₂
Na ₃ V ₂ (PO ₄) ₃ /C ⁷²	Zn ²⁺ , Na ⁺ insertion/extraction	0.8–1.7 V	1.37	93 mA h g ⁻¹ (0.5 C)	0.5 M CH ₃ COONa, 1 M Zn(CH ₃ COO) ₂
Na ₃ V ₂ (PO ₄) ₂ F ₃ ⁴²	Zn ²⁺ insertion/extraction	0.8–1.9 V	1.60 and 1.35	64.7 mA h g ⁻¹ (0.08 A g ⁻¹)	2 M Zn(CH ₃ F ₃ SO ₃) ₂
Li ₃ V ₂ (PO ₄) ₃ ⁷³	Zn ²⁺ insertion/extraction	0.7–2.1 V	1.72, 1.32, 1.24	128 mA h g ⁻¹ (0.2 C)	1 M Li ₂ SO ₄ , 2 M ZnSO ₄
Na ₃ V ₂ (PO ₄) ₃ ⁷³	Zn ²⁺ insertion/extraction	0.7–2.1 V	1.35	96 mA h g ⁻¹ (0.2 C)	1 M Li ₂ SO ₄ , 2 M ZnSO ₄
LiMn _{0.8} Fe _{0.2} PO ₄ ⁷⁴	Zn ²⁺ insertion/extraction	1.0–2.35 V	1.3 and 1.4	144 mA h g ⁻¹ (0.1 C)	21 m LiTFSI, 0.5 M ZnSO ₄
KCu[Fe ³⁺ (CN) ₆] ⁷¹	Na ⁺ insertion/extraction	1.4–2.1 V	1.6	61 mA h g ⁻¹ (5 C)	1 M ZnSO ₄ , 0.01 M H ₂ SO ₄
Na ₂ MnFe(CN) ₆ ⁹²	Na ⁺ insertion/extraction	0.2–1.2 V	1.7 and 1.5	140 mA h g ⁻¹ (0.5 C)	1 M Na ₂ SO ₄ , 1 M ZnSO ₄

Table 1. continued

cathode material	reaction mechanism	testing voltage	main operating voltage (V)	discharge capacity	electrolyte components
KCuFe(CN) ₆ ³⁷	Zn ²⁺ insertion/extraction	—	0.6 V vs SCE	56 mA h g ⁻¹ (20 mA g ⁻¹)	1 M ZnSO ₄
FeFe(CN) ₆ ⁶⁸	Zn ²⁺ insertion/extraction	1.0–2.0 V	1.35	112 mA h g ⁻¹ (10 mA g ⁻¹)	1.0 M Zn(OAc) ₂ in [Ch]OAc + water
KCu[Fe ³⁺ (CN) ₆] ⁶⁹	Zn ²⁺ insertion/extraction	0.45–1.4 V vs SCE	0.7 and 0.9 vs SCE	<55 mA h g ⁻¹ (60 mA g ⁻¹)	20 mM ZnSO ₄
Zn ₃ [Fe(CN) ₆] ₂ ³⁶	Zn ²⁺ insertion/extraction	0.8–2.0 V	1.57	65.4 mA h g ⁻¹ (60 mA g ⁻¹)	1 M ZnSO ₄
ZnMo ₆ S ₈ ⁴³	Zn ²⁺ insertion/extraction	0.25–1.0 V	0.5 and 0.35	135 mA h g ⁻¹ (6.4 mA g ⁻¹)	0.1 M ZnSO ₄
Mo ₆ S ₈ ⁴⁴	Zn ²⁺ insertion/extraction	0.25–1.0 V	0.5 and 0.34	120 mA h g ⁻¹ (0.1 C)	1 M ZnSO ₄
quinone(C4Q) ⁴⁵	Zn ²⁺ insertion/extraction	0.8–1.3 V	1.0	333 mA h g ⁻¹ (50 mA g ⁻¹)	3 M Zn(CF ₃ SO ₃) ₂
p-chloranil ⁴⁶	Zn ²⁺ insertion/extraction	0.8–1.4 V	1.1	≥200 mA h g ⁻¹ (217 mA g ⁻¹)	1 M Zn(OTf) ₂ ·H ₂ O
Co(III) rich-Co ₃ O ₄ ⁷⁵	conversion reaction	0.8–2.2 V	2.0	158 mA h g ⁻¹ (1 A g ⁻¹)	2 M ZnSO ₄ , 0.2 M CoSO ₄

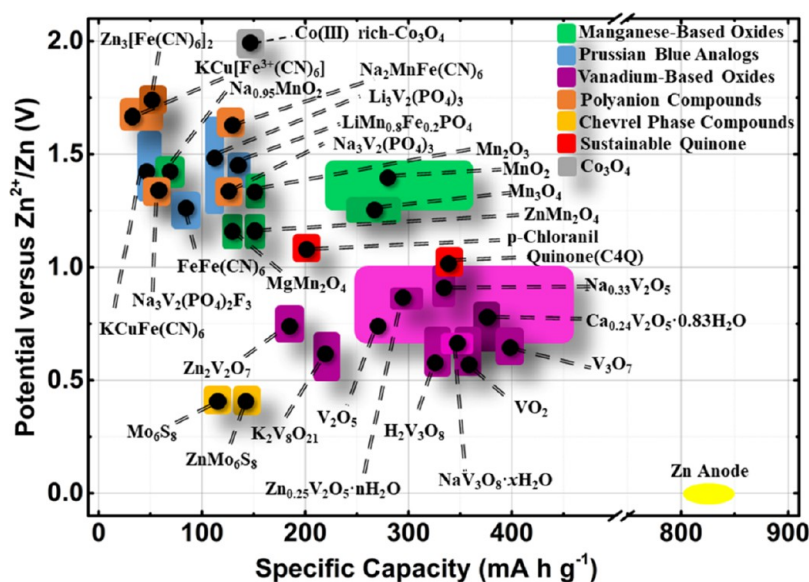


Figure 5. Operating voltage versus specific capacity for Zn anode and various cathode materials presently used for aqueous ZIBs.

High operating voltage is desirable to improve the energy density of the battery. Other PBAs such as copper hexacyanoferrate^{37,68,69} also exhibited a high operating voltage, for instance, a cathodic peak at ~ 0.6 V and a corresponding anodic peak at ~ 0.8 V vs SHE (Figure 8C), with a reversible insertion/extraction of Zn²⁺ ions into/from the A sites of PBAs crystal lattice structure. However, the serious issue is that most of these materials deliver an extremely low capacity of less than 70 mA h g⁻¹ (Figure 8D), leading to insufficient energy density, which greatly restricts their further development.

Vanadium-Based Oxides. Owing to the low cost, abundant resources, and multivalence of vanadium, vanadium oxides and their derivatives are the most important compounds being used in the field of battery systems, particularly in LIBs and SIBs.^{93–99} In recent years, research has been undertaken for aqueous ZIBs as well, wherein metal vanadates were tested as a promising cathode. Nazar et al. constructed a layered vanadium oxide bronze (Zn_{0.25}V₂O₅·nH₂O),³⁸ in which interlayer metal ions (Zn²⁺) and structural water act as pillars to stabilize the structure during cycling (Figure 9A). The Zn/Zn_{0.25}V₂O₅·nH₂O cell delivered a very high capacity of 282 mA h g⁻¹ at 300 mA h g⁻¹ with an average operating voltage of ~ 0.9 V (Figure 9B). This means the cell can display a specific energy of ~ 250 Wh kg⁻¹ (cathode only). The structural water

molecules are revealed to play a crucial role by reversibly expanding and contracting the interlayer spacing of Zn_{0.25}V₂O₅ that facilitates Zn²⁺ ingress/egress, thus resulting in good kinetics and high rate performance.³⁸ The effect of structural water on Zn²⁺ storage performance is further demonstrated in hydrated V₂O₅·nH₂O reported by Mai and Yang.³⁹ The solid-state magic-angle-spinning (MAS) ¹H NMR is conducted to explore structural evolution and the effect of structural water in the electrode materials (Figure 9C). The ¹H spectrum of pristine hydrated V₂O₅·nH₂O (black line) shows the broad resonance at 5.6 ppm, which is attributed to the structural water trapped between V₂O₅ bilayers. After charging to 1.3 V, ¹H peak (blue lines) shifts slightly upward to 5.3 ppm with an increased signal intensity accompanied with the decrease in the V₂O₅·nH₂O interlayer distance from 12.6 to 10.4 Å. As evidenced by the authors, this can be explained by the formation of hydrogen bonds among Zn²⁺, CF₃SO₃⁻, H₂O, and lattice oxygen that pull V₂O₅·nH₂O bilayers closer. After discharging to 0.2 V, ¹H peak (red lines) splits into two peaks. One is a relatively sharp peak at 5.1 ppm that is attributed to water molecules in the few remaining clusters of Zn²⁺, CF₃SO₃⁻, and H₂O, while a relatively broad peak at 2.9 ppm is ascribed to the water molecules associated with the inserted Zn²⁺, thus increasing bilayer distance from 10.4 to 13.5 Å (Figure 9C).

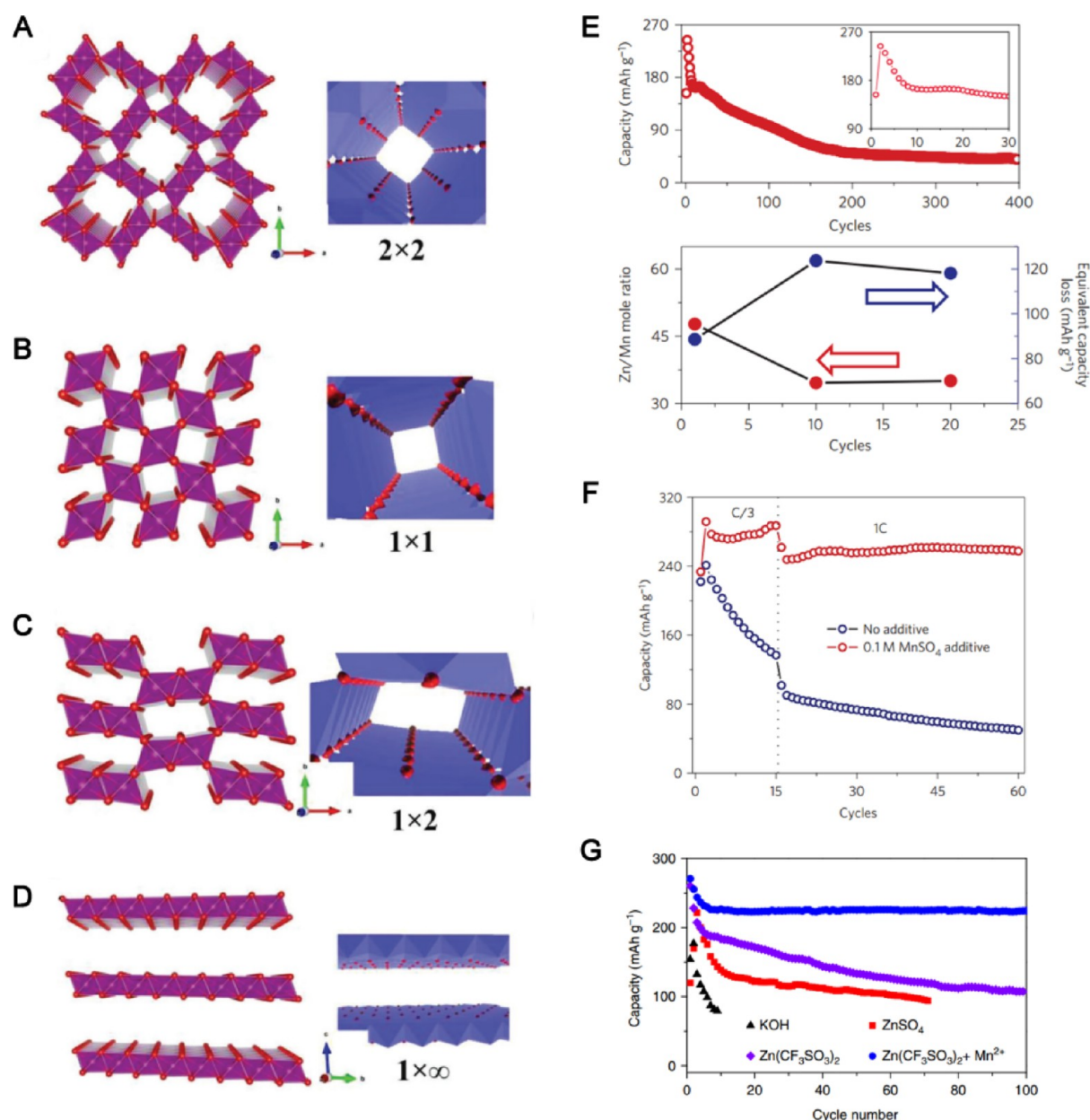


Figure 6. Schematic diagram of various crystal structures of MnO₂: (A) α-MnO₂, (B) β-MnO₂, (C) γ-MnO₂, and (D) δ-MnO₂. Reprinted with permission from ref 85. Copyright 2018 Wiley-VCH. (E) Cycling performance at C/5 (1C = 308 mA h g⁻¹) and elemental analysis of dissolved Mn²⁺ ions in the 2 M ZnSO₄ aqueous electrolyte during cycling. (F) Cycling performance of α-MnO₂ electrodes with and without 0.1 M MnSO₄ additive in a 2 M ZnSO₄ aqueous electrolyte. Reprinted with permission from ref 50. Copyright 2016 Macmillan Publishers. (G) Comparison of the cycling performance of β-MnO₂ with electrolytes of 3 M Zn(CF₃SO₃)₂, and 3 M Zn(CF₃SO₃)₂ with 0.1 M Mn(CF₃SO₃)₂ additive. Reprinted with permission from ref 56. Copyright 2017 Macmillan Publishers.

It seems that these water molecules effectively function like a “lubricant” to facilitate fast Zn²⁺ transportation. Moreover, the charge shield effect of the crystal H₂O molecules could reduce the effective charges of inserted Zn²⁺ ions and thus enhance the capacity and rate performance.

Except for structural water, electrolyte optimization may also be a promising method to improve the electrochemical performance for vanadium-based cathodes. Chen et al. found that aqueous ZnSO₄ electrolyte could lead to the fast dissolution of NaV₃O₈·1.5H₂O as well as the formation of vertical and harsh Zn dendrites, thus resulting in a rapid degradation in capacity (Figure 10A).⁵² The addition of Na⁺ into the electrolyte can change the dissolution equilibrium of Na⁺ from

NaV₃O₈·1.5H₂O electrodes and thus impede the continuous dissolution of NaV₃O₈·1.5H₂O, as supported by the observation that the electrolyte (1 M ZnSO₄ and 1 M Na₂SO₄) is transparent and colorless even when the NaV₃O₈·1.5H₂O electrode is dipped into the electrolyte for 24 h (inset in Figure 10A), indicating that the dissolution of NaV₃O₈·1.5H₂O was suppressed. As a result, the capacity of a Zn//NaV₃O₈·1.5H₂O battery with 1 M Na₂SO₄ additive can stabilize at 221 mA h g⁻¹ after 100 cycles with a retention rate of 90% at a current density of 1 A g⁻¹ (Figure 10B).

In addition, a series of metal vanadates with or without structural water, e.g., Zn₃V₂O₇(OH)₂·2H₂O,⁴⁰ Ca_{0.25}V₂O₅·nH₂O,⁵⁹ LiV₃O₈,¹⁰⁰ NaV₃O₈·1.5H₂O,⁵² Na₂V₆O₁₆·

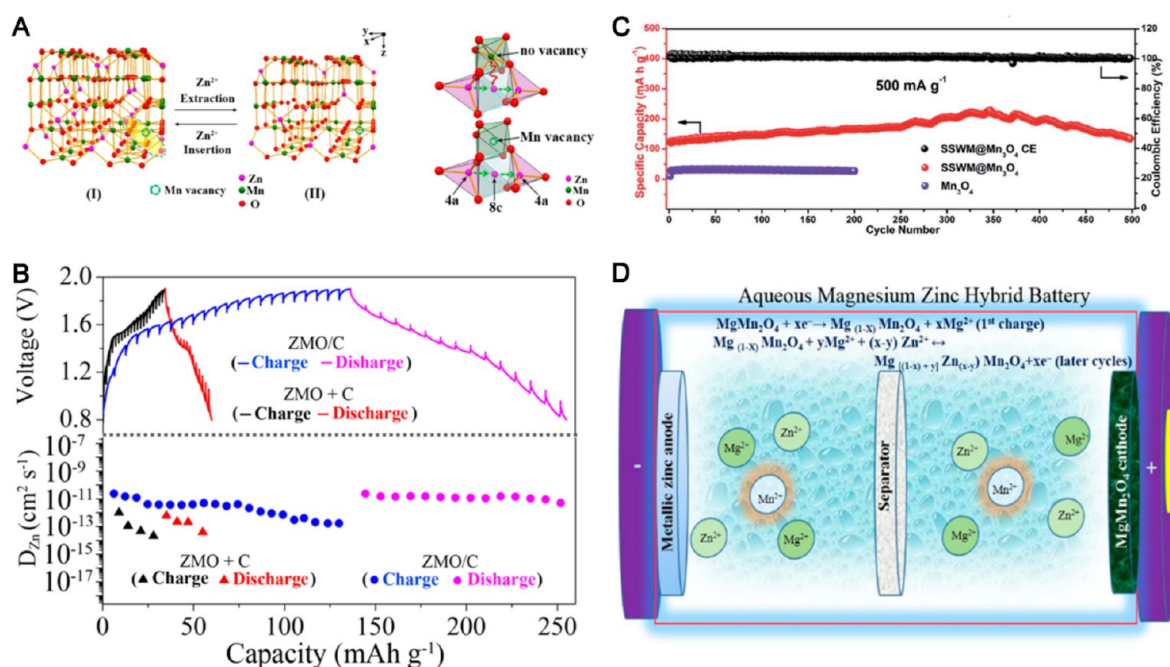


Figure 7. (A) Schematic illustration of Zn²⁺ insertion/extraction in a spinel ZnMn₂O₄ framework and proposed Zn²⁺ diffusion pathway in ZnMn₂O₄ framework without and with Mn vacancies. (B) Galvanostatic intermittent titration technique profiles and calculated Zn²⁺ diffusion coefficient of ZnMn₂O₄/C and ZnMn₂O₄+C electrodes. Reprinted from ref 33. Copyright 2016 American Chemical Society. (C) Long-term cycling performance of Mn₃O₄ with or without 3D conductive substrates. Reprinted with permission from ref 84. Copyright 2018 The Royal Society of Chemistry. (D) An aqueous magnesium zinc hybrid battery. Reprinted from ref 88. Copyright 2018 American Chemical Society.

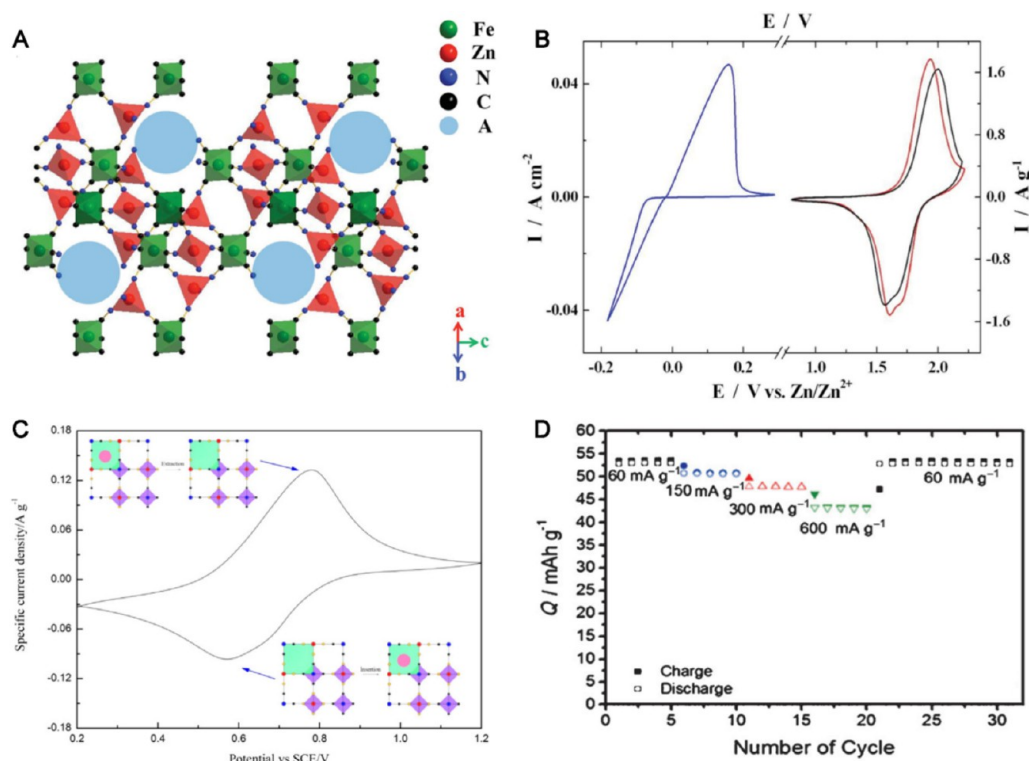


Figure 8. (A) Crystal structure of rhombohedral zinc hexacyanoferrate. (B) CV curves at a scan rate of 2 mV s⁻¹ of Zn anode (blue), zinc hexacyanoferrate (red), and full cell (black). Reprinted with permission from ref 36. Copyright 2015 Wiley-VCH. (C) Typical cyclic voltammograms of copper hexacyanoferrate at 1 mV s⁻¹. Reprinted with permission from ref 37. Copyright 2015 Elsevier. (D) Rate capability of copper hexacyanoferrate. Reprinted with permission from ref 69. Copyright 2015 Wiley-VCH.

1.63H₂O,^{60,61} Na_{1.1}V₃O_{7.9},¹⁰¹ Na_{0.33}V₂O₅,⁶² Na₅V₁₂O₃₂,⁶⁴
K₂V₈O₂₁,⁶⁵ Zn₂V₂O₇,¹⁰² Mo_{2.5+y}VO_{9+z},¹⁰³ Li_xV₂O₅·nH₂O,¹⁰⁴

and Ag_{0.4}V₂O₅,¹⁰⁵ are developed as cathode materials for aqueous ZIBs because of their layered or tunnel structure.

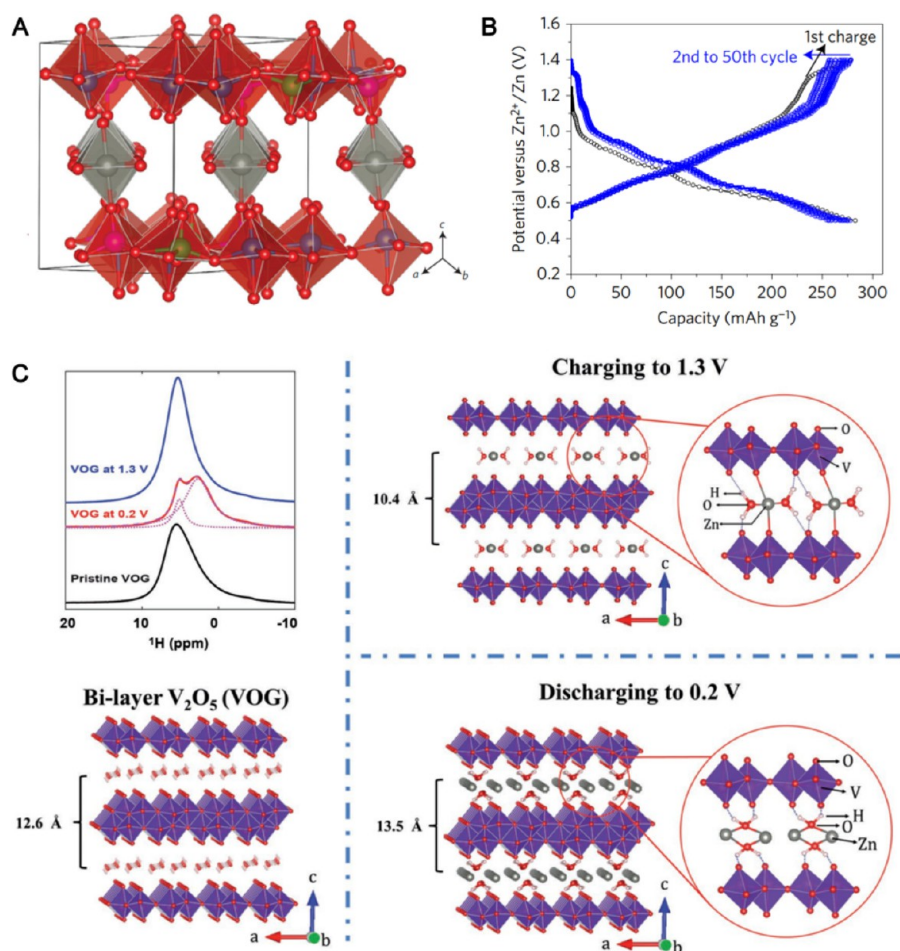


Figure 9. (A) Crystal structure and (B) galvanostatic discharge and charge profiles at 300 mA g^{-1} of $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$. Reprinted with permission from ref 38. Copyright 2016 Macmillan Publishers. (C) Ex situ solid state ^1H NMR and the proposed crystal structures of pristine $\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ and $\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ after charging to 1.3 V and discharging to 0.2 V. Reprinted with permission from ref 39. Copyright 2018 Wiley-VCH.

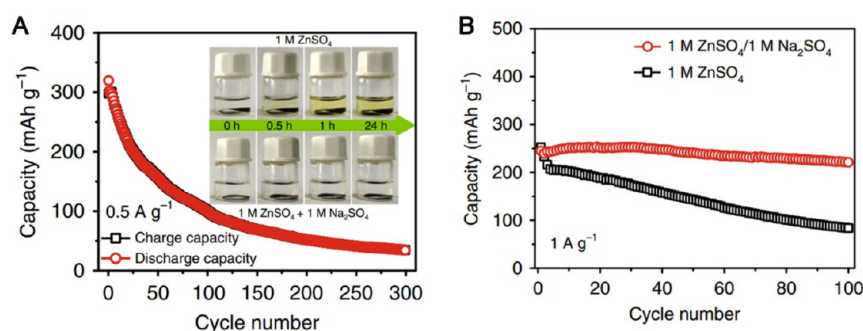


Figure 10. (A) Cycling performance of $\text{NaV}_3\text{O}_8 \cdot 1.5\text{H}_2\text{O}$ electrode in ZnSO_4 electrolyte. The insets are optical images of electrodes in ZnSO_4 and $\text{ZnSO}_4/\text{Na}_2\text{SO}_4$ electrolytes for different periods. (B) Cycling performance (1 A g^{-1}) of $\text{NaV}_3\text{O}_8 \cdot 1.5\text{H}_2\text{O}$ electrodes in ZnSO_4 and $\text{ZnSO}_4/\text{Na}_2\text{SO}_4$ electrolytes. Reprinted with permission from ref 52. Copyright 2018 Macmillan Publishers.

In the meantime, various vanadium oxides, such as V_2O_5 ,^{58,106,107} VO_2 ,¹⁰⁸ $\text{V}_3\text{O}_7 \cdot \text{H}_2\text{O}$ ($\text{H}_2\text{V}_3\text{O}_8$),^{109–111} and so on, are also investigated as cathodes for aqueous ZIBs. Most of them show good electrochemical performance in terms of high specific capacity, good rate capability, high energy density, and long cycle life. As a typical example, an aqueous $\text{Zn}-\text{V}_2\text{O}_5$ battery reported by Zhang et al. exhibits the highest energy density at high power (322 Wh kg^{-1} at 710 W kg^{-1} ; 274 Wh kg^{-1} at 7100 W kg^{-1}) among the reported vanadium-based cathodes.⁵⁸ It also demonstrates a long-term

cycling stability with a high reversible capacity of 372 mA h g^{-1} (ca. 91.1% capacity retention) even after 4000 cycles.

Despite their promising performance, however, most of them suffer from strong electrostatic interactions between the divalent Zn^{2+} ions and the host lattice, which can cause slow Zn^{2+} diffusion and trap a certain amount of Zn^{2+} ions in host lattice or form a zinc-containing vanadium phase.^{58,60,61,100,106} It is therefore necessary to develop new synthetic strategies for vanadium-based materials that reversibly insert Zn^{2+} ions. Motivated by the observation that the cation-deficient anatase

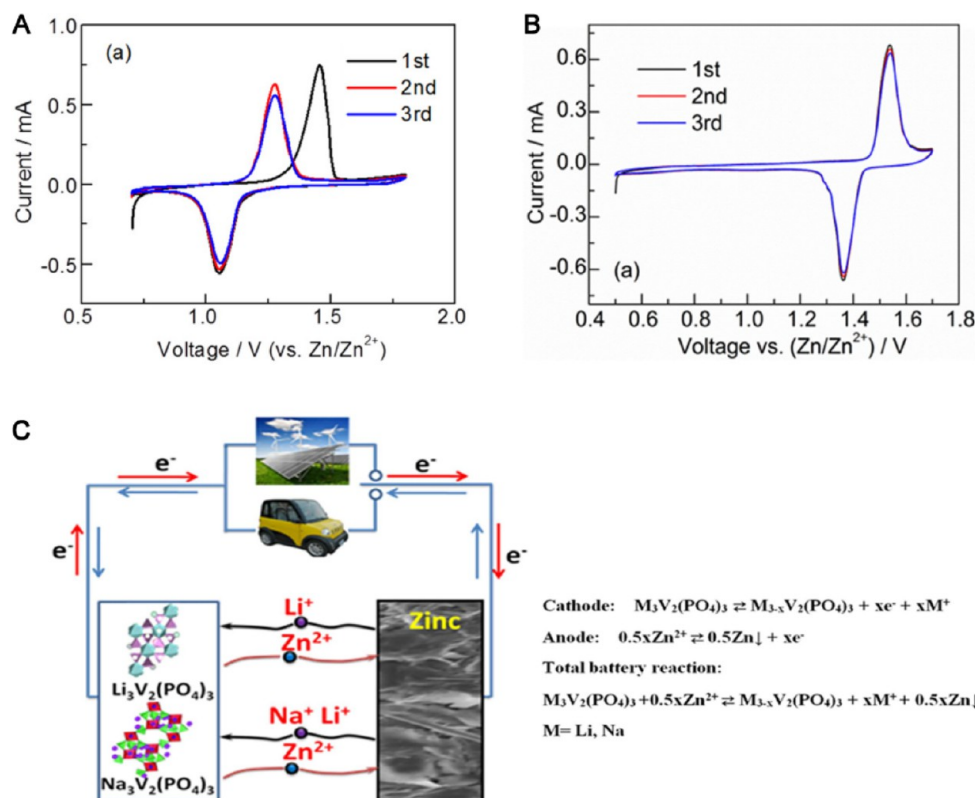


Figure 11. (A) CV behaviors of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ at $0.5 \text{ mol L}^{-1} \text{ Zn}(\text{CH}_3\text{COO})_2$. Reprinted with permission from ref 41. Copyright 2016 Elsevier. (B) CV behaviors of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ at $0.5 \text{ mol L}^{-1} \text{ CH}_3\text{COONa}/\text{Zn}(\text{CH}_3\text{COO})_2$. Reprinted with permission from ref 72. Copyright 2016 Elsevier. (C) The operation scheme of $\text{M}_3\text{V}_2(\text{PO}_4)_3$ in a hybrid aqueous electrolyte ($1 \text{ M Li}_2\text{SO}_4 + 2 \text{ M ZnSO}_4$). Reprinted with permission from ref 73. Copyright 2016 Macmillan Publishers.

TiO_2 exhibits high electrochemical activity for multivalent MIBs and AIBs,¹¹² the possibility of using the structural defects for aqueous ZIBs is encouraged. In fact, as mentioned earlier, cationic defects have been applied in manganese-based materials (ZnMn_2O_4), in which Mn vacancies enable fast diffusion of Zn^{2+} without much electrostatic barrier.³³ Therefore, the chemistry of defects may also be applicable to vanadium-based materials. In addition, because of the high electronegative oxygen that can easily bond with the divalent Zn^{2+} ions,²⁰ the vanadium-based compounds without oxygen, such as vanadium sulfides, vanadium nitrides, and so on, could be considered because the sulfur or nitrogen owns lower electronegativity than that of oxygen.

Polyanion Compounds. In the past decade, extensive research efforts have been focused on polyanion compounds as cathode materials in both aqueous and nonaqueous LIBs or SIBs because of their open channels for diffusion of Li^+ and Na^+ .^{113–115} Recently, polyanion compounds have been applied in aqueous ZIBs as reported in several contributions, wherein NASICON-type $\text{M}_3\text{V}_2(\text{PO}_4)_3$ ($\text{M} = \text{Li, Na}$) have attracted considerable interest because of their thermodynamically stable structure.^{41,72,73} Although the radius of the Zn^{2+} ion is $0.74 \text{ \AA}$, which is smaller than that of the Na^+ ion ($0.99 \text{ \AA}$), the cation radius would not be the only factor under consideration during the insertion of Zn^{2+} in aqueous ZIBs because the high polarization of the Zn^{2+} ion may limit its migration dynamics.⁴¹

Huang et al. developed a NASICON structured $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ as cathode material for aqueous ZIBs using $0.5 \text{ mol L}^{-1} \text{ Zn}(\text{CH}_3\text{COO})_2$. They found a large voltage difference between the first and second charge curves for $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ cathode, as

shown in Figure 11A, which is confirmed to be derived from the difference of chemical potential between $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Zn}_x\text{Na}_{3-x}\text{V}_2(\text{PO}_4)_3$. The first oxidation peak at about 1.46 V corresponds to Na^+ deintercalation from $\text{Na}_3\text{V}_2(\text{PO}_4)_3$. However, the redox pair peaks during subsequent cycles appear at around 1.05 and 1.28 V , which can be attributed to the deintercalation/intercalation of Zn^{2+} from/into the $\text{NaV}_2(\text{PO}_4)_3$ host. This demonstrated that during the discharge process, a Zn-inserted phase $\text{Zn}_x\text{Na}_{3-x}\text{V}_2(\text{PO}_4)_3$, which has a redox potential lower than that of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$. Preadding the Na^+ could suppress the formation of $\text{Zn}_x\text{Na}_{3-x}\text{V}_2(\text{PO}_4)_3$ phase, as demonstrated by Huang's further work.⁷² The $\text{Zn}/\text{Na}_3\text{V}_2(\text{PO}_4)_3$ cell with a hybrid aqueous electrolyte of $0.5 \text{ mol L}^{-1} \text{ CH}_3\text{COONa}/\text{Zn}(\text{CH}_3\text{COO})_2$ exhibits almost overlapped cyclic voltammetry (CV) curves for the initial three cycles with a redox couple at about 1.37 and 1.52 V (Figure 11B), demonstrating the excellent structural reversibility of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$. This electrochemical behavior for NASICON-type compounds is further verified by Zhao's work,⁷³ which exhibits only Li^+ insertion/extraction into/from both the $\text{Li}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ cathodes in a hybrid aqueous electrolyte ($1 \text{ M Li}_2\text{SO}_4/2 \text{ M ZnSO}_4$) (Figure 11C). Nonetheless, more evidence is required to determine how Zn^{2+} is suppressed by Li^+ or Na^+ to enter the host of the NASICON-type structure.

Chevrete Phase Compounds. Another class of cathode materials for aqueous ZIBs is the Chevrete phase compounds with the general formula $\text{M}_x\text{Mo}_6\text{T}_8$ ($\text{M} = \text{metal}$; $\text{T} = \text{S, Se, or Te}$),^{43,44} in which six molybdenum atoms form an octahedral Mo_6 cluster, surrounded by a cubic unit (T_8) of chalcogenide atoms. Six of the T_8 are also bonded axially to a molybdenum

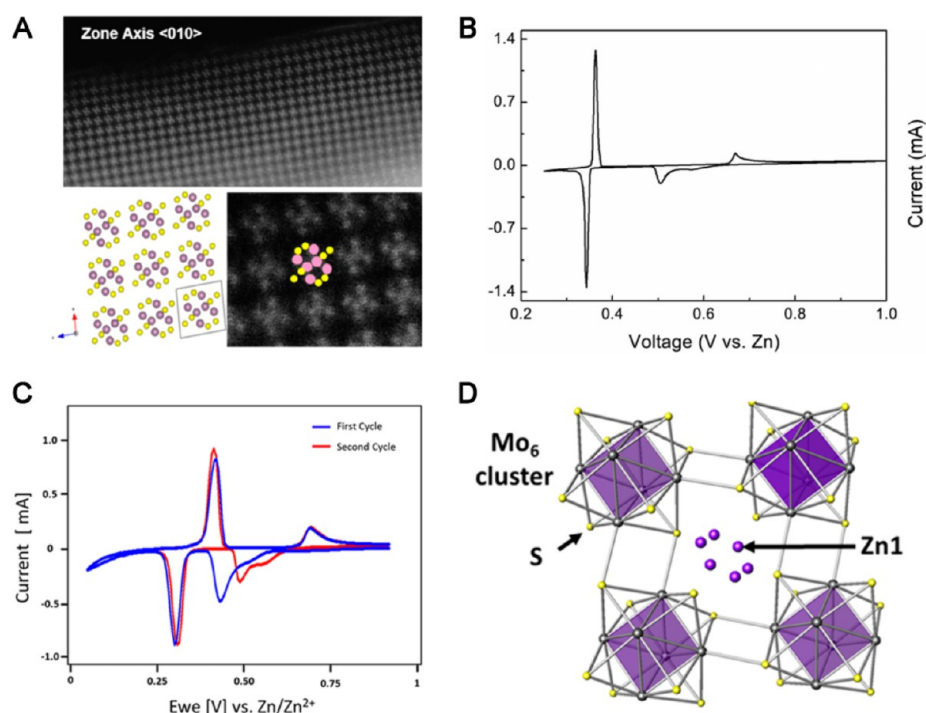


Figure 12. (A) Schematic crystal structure of Mo₆S₈ phase and corresponding STEM image. (B) CV curve of Mo₆S₈ for intercalating Zn²⁺ ions in 1.0 M ZnSO₄ in water. Reprinted from ref 44. Copyright 2016 American Chemical Society. (C) CV curve of the Mo₆S₈ electrode at 0.05 mV s⁻¹ in a 0.1 M ZnSO₄ aqueous electrolyte. (D) Local structure around the Zn1 positions in Zn_xMo₆S₈. Reprinted from ref 43. Copyright 2016 American Chemical Society.

atom of a neighboring cluster, resulting in the three-dimensional network crystal structure creating cavities surrounded by sulfur atoms (Figure 12A). Chevrel phase compounds have outstanding capability to host various cations (Li⁺, Na⁺, Mg²⁺, Al³⁺, etc.) with remarkable ionic and electronic transportation kinetics.^{116–118}

Recently, Chevrel phase Mo₆S₈ has shown good potential for Zn²⁺ ion intercalation in aqueous ZIBs.⁴⁴ As depicted in Figure 12B, CV curves had two sets of redox peaks including two anodic peaks at 0.68 and 0.37 V and two cathodic peaks at 0.50 and 0.34 V. It is worth noting that this phase exhibited some irreversibility by comparing the first and second intercalation cycles as its second pair peak appeared at higher voltage (Figure 12C), which is presumably due to Zn²⁺ ion trapping in the lattice of Mo₆S₈ to form Zn_xMo₆S₈ phase (0 ≤ x ≤ 2).¹¹⁷ It is reported that the Zn²⁺ ions in Zn_xMo₆S₈ were statistically distributed with a partial occupancy of 1/6, meaning that the Zn1 positions generated by the symmetry formed a six-membered ring (Figure 12D).⁴³ In spite of its outstanding cyclic stability after initial cycle, the low redox potentials and low capacity make them unsuitable for cathode materials, but they can be considered as anode materials.⁴⁴

Sustainable Quinone Cathodes. To date, great efforts have been made on building high-performance ZIBs using inorganic compounds and Prussian blue analogs, as discussed above. Because they are lightweight, environmentally friendly, and sustainable, organic compounds are also attractive for electrochemical energy storage devices;^{119,120} especially, they have shown great superiority in both monovalent (Li⁺ and Na⁺) and divalent (Mg²⁺) cation storage.^{121–124} Among them, quinones, which are ubiquitous in nature, can act as sustainable and green electrode materials. However, currently most battery systems using quinone electrodes work with organic electrolytes,

which suffer from the fast capacity fading due to the dissolution of active materials.¹¹⁹ Applying quinone electrodes in aqueous batteries should be a smart choice as they can stably exist in water, which have been proven to have versatile properties in aqueous (H⁺, Li⁺, Na⁺, K⁺, and Mg²⁺) rechargeable batteries.¹²⁵

Very recently, Chen and co-workers reported a series of quinone compounds as high-performance cathodes in aqueous ZIBs, in which the calix[4]quinone (C4Q) with open bowl structures and eight carbonyls can provide reversible and efficient Zn storage.⁴⁵ The C4Q cathode displays a flat and safe operation voltage of 1.0 V with a low polarization of 70 mV, which contributes to the stable energy output, and thus acquires a capacity of 335 mA h g⁻¹, corresponding to uptake of three Zn ions (Zn₃C4Q) and utilization of six carbonyls. The electrostatic potential (ESP) method was applied to deduce the active sites for Zn²⁺ storage in C4Q. It is known that the sites with more positive ESP tend to occur in nucleophilic reactions, whereas the more negative ESP areas prefer electrophilic reactions.⁴⁵ In other words, the sites with more negative ESP are attractive for discharge reaction as it involves Zn²⁺ uptake. For the C4Q molecular structure, the carbonyls on and under a molecule display lower ESP than the bilateral carbonyls, suggesting that the Zn²⁺ ions have strong interaction with two carbonyls at the top and four carbonyls at the bottom. Therefore, the redox reaction of Zn²⁺ uptake in C4Q is as follows:



Other quinone-like organic compounds such as tetrachloro-1,4-benzoquinone (also called p-chloranil) are also explored as a potential cathode for aqueous ZIBs.⁴⁶ The CV curve of p-chloranil cathode features only one pair of redox peaks, indicating a single-step reduction/oxidation of p-chloranil (Figure 13A).

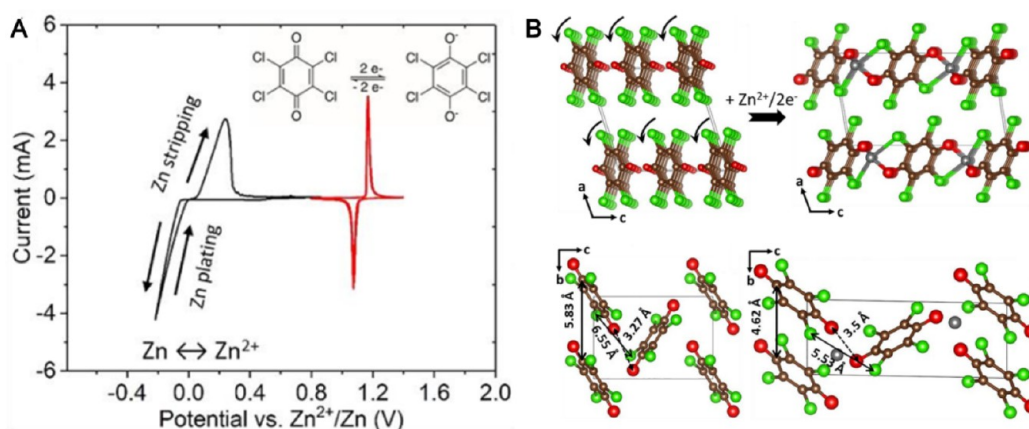


Figure 13. (A) CV curve of the Zn anode (black) against the p-chloranil cathode (red) in 1 M $\text{Zn}(\text{OTf})_2\text{-H}_2\text{O}$ electrolyte. (B) Structural models of p-chloranil and Zn-p-chloranil ($\text{Zn}\cdot\text{C}_6\text{Cl}_4\text{O}_2$) as obtained from DFT structural optimization. Reprinted from ref 46. Copyright 2018 American Chemical Society.

It has an average working voltage of 1.1 V with a very small voltage polarization of 50 mV. The first-principles density functional calculations show that the p-chloranil molecules undergo a squeezing reorientation upon insertion of Zn^{2+} , which increases the O–O distance and decreases the Cl–Cl distance compared to chloranil (Figure 13B). As a result, p-chloranil delivers a high capacity of $\geq 200 \text{ mA h g}^{-1}$ in a flat plateau around 1.1 V, which equate to an attractive specific energy of $>200 \text{ Wh kg}^{-1}$ at an unparalleled energy efficiency ($\sim 95\%$).

Although the raw quinone compounds are scarcely soluble in water, it should be considered for long-life cycling that the slight solubility of discharge products (e.g., $\text{Zn}_x\text{C}_4\text{Q}$) is accompanied by its possible side reaction on the zinc anode. That is, the dissolved C_4Q^{2-} in the electrolyte can pass through the membrane and react with zinc, forming a byproduct on the anode side, which causes the decline of capacity.⁴⁵ To overcome this issue, a cation-exchange membrane (Nafion) was introduced as a separator to suppress the dissolution of the discharge product and protect the zinc anode from passivation by discharged products, thus leading to excellent cyclic performance. Although these quinones show great promise in aqueous ZIBs, they are still in their infancy, which requires the development of more categories by elaborately designing the molecular structure to achieve higher capacity and voltage in further studies.

Zn Anode in Aqueous ZIBs. Metallic zinc (Zn) has been regarded as an ideal anode material for aqueous batteries because it features high abundance and low cost (ca. USD $\$2 \text{ kg}^{-1}$),⁴⁰ low flammability and toxicity, high electrical conductivity, easy processing, and high compatibility and stability in water.⁴¹ Furthermore, the Zn anode possesses a suitable redox potential (-0.76 V vs standard hydrogen electrode (SHE)) in aqueous electrolyte,³⁶ a high capacity (ca. 820 mA h g^{-1} ; $5851 \text{ mA h cm}^{-3}$),⁵⁹ and low polarizability.²⁵ Early attempts in alkaline electrolyte, however, are somewhat far from practical application owing to the following issues: (1) low plating/stripping Coulombic efficiency of Zn anode;³¹ (2) growth of Zn dendrite during cycling;²⁹ (3) sustained consumption of electrolyte;²³ and (4) formation of irreversible byproducts (e.g., ZnO or $\text{Zn}(\text{OH})_2$) on the Zn anode,¹²⁶ resulting in severe capacity fading and poor Coulombic efficiency of cell. These drawbacks in alkaline electrolyte system directly drive the recent progress of aqueous ZIBs using neutral or mild acidic aqueous electrolytes.

Although dendritic zinc could be minimized in mild neutral pH (or slightly acidic) electrolyte,⁵⁹ the poor reversibility and

low plating/stripping Coulombic efficiency (CE) of Zn anode still challenge its practical application. The main problems associated with the low CE are the consumption of Zn and electrolyte by side reactions, as in the case of the lithium metal anode.¹²⁷ Therefore, significant excessive zinc needs to be used to maintain the cycle stability, but this leads to suboptimal utilization of its theoretical capacity. In addition, the regular replenishing of the electrolyte with water was often required to compensate the water decomposition.^{23,128} Despite most previous reports that high charge/discharge rates were used to minimize the effect of their poor reversibility,^{36,72,129} the goal of achieving high CE in aqueous ZIBs remains a challenge.

To address these issues, many research efforts have been focused on electrolyte optimization; these include the species and concentration of zinc salts, as well as the additive in electrolyte,^{31–33,52,106} which will be discussed later. It is known that the formation of zinc dendrite and corrosion mainly originate from uneven Zn^{2+} distribution on the flat or two-dimensional metallic Zn foil. The sharp end of the initial small dendrite tip on traditional planar current collectors¹³⁰ could serve as a charge center in the electric field for continuous accumulation of charges, thus amplifying the growth of dendrites via further deposition on those sharp ends. In contrast, the hierarchical current collector with high electroactive surface area and uniform electric field could inhibit dendrite formation.¹³¹ Therefore, deposition of Zn into hierarchical current collectors may represent a potential strategy to alleviate dendrite growth and achieve high rate and stable ZIBs. For instance, Fan et al. designed a Zn nanoflakes array-coated 3D graphene foam (Zn-GF) anode (Figure 14A),¹³² in which 3D porous GF acts as a highly conductive substrate and macroporous agent to alleviate the formation of inactive surface oxide or hydroxide layer and dendrite growth. Moreover, this Zn-GF anode possesses higher capacity and smaller polarization (Figure 14B), as well as lower charge-transfer resistance (Figure 14C) than that of conventional compact Zn foil, implying the improved electrochemical activity and fast Zn/Zn^{2+} redox kinetics of the zinc nanoarray structure. In addition, several significant works also focused on this field, such as Zn-coated carbon cloth anode (Figure 14D),¹³³ Zn-coated carbon nanotube paper anode (Figure 14E),¹³⁴ Zn-coated carbon nanotube yarns (Figure 14F),¹³⁵ Zn-coated nitinol yarns,¹³⁶ etc. These novel Zn-hierarchical current collector anodes greatly diminish the formation of dendrites

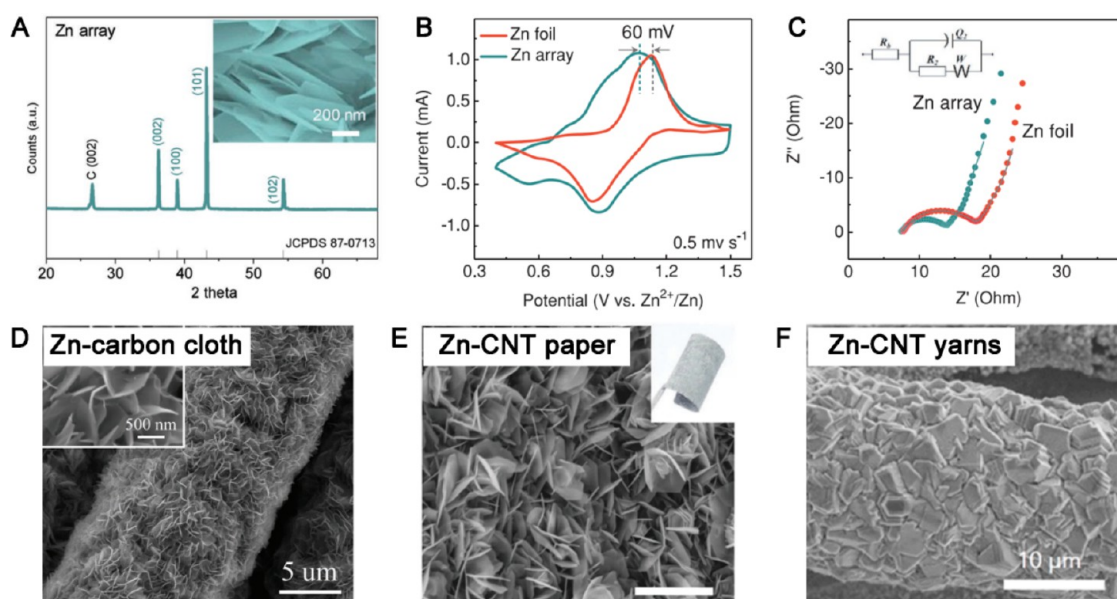


Figure 14. (A) XRD pattern of the electrodeposited Zn array-coated graphene foam. Inset: SEM image showing the Zn flake-array structure. (B) CV curve comparison after 5 cycles charge–discharge activation at 0.5 C of the quasi-solid-state ZIBs made of zinc orthovanadate cathode and Zn array or foil anode. (C) EIS of the Zn array and Zn foil electrodes at the fully charged state. Reprinted with permission from ref 137. Copyright 2018 Wiley-VCH. (D) SEM images of Zn-coated carbon cloth anode. Reprinted with permission from ref 133. Copyright 2017 Wiley-VCH. (E) SEM image of Zn-coated carbon nanotube paper anode. Reprinted with permission from ref 134. Copyright 2018 The Royal Society of Chemistry. (F) SEM image of Zn-coated carbon nanotube yarns. Reprinted from ref 135. Copyright 2018 American Chemical Society.

and are favorable to enhanced high rate capability and long durability.

Aqueous Electrolytes. Electrolyte plays a vital role in battery chemistry as its functions to transport guest ions and make a connection between cathode and anode. The use of aqueous electrolytes offers many advantages in terms of low cost, operational safety facile manufacture, environmental benignity and high ionic conductivities.⁷ Currently, the ZnSO_4 or $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ salt-based electrolytes are considered to be promising electrolytes and widely used in aqueous ZIBs, as shown in Table 1. Many works focus on adjusting the concentration of zinc salts or using the additive to encourage better electrochemical performance for aqueous ZIBs. Chen et al. investigated four mild acidic solutions separately containing zinc salt of ZnCl_2 , ZnNO_3 , ZnSO_4 , and $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ in a $\text{Zn}/\text{ZnMn}_2\text{O}_4$ system.³³ The voltammetry in $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ electrolyte (Figure 15A) and ZnSO_4 electrolyte (Figure 15B) indicates good reversibility of Zn stripping/plating for both electrolytes. It also demonstrates a poor CE for ZnCl_2 and ZnNO_3 electrolytes associated with the instability of Cl^- and NO_3^- , which is also observed in a $\text{Zn}-\text{V}_2\text{O}_5$ system by Zhou and co-workers.¹⁰⁶ The effect of zinc salt concentration on Zn stripping/plating CE was also discussed by Chen et al., who indicated that a high-concentration electrolyte with increased viscosity (Figure 15C) displays better CE than that of low-concentration counterparts. It is supposed that higher salt concentration could not only alter the solvation and transporting behaviors of cations/anions to enhance stability and kinetics¹³⁸ but also reduce the water activity and water-induced side reactions.¹³⁹ The CE of the $\text{Zn}/\text{ZnMn}_2\text{O}_4$ cell increases with salt concentration and is almost unchanged ($\sim 100\%$) at 3 M (Figure 15D), which enables long-term stability with a capacity retention of 94% over 500 cycles at 500 mA g^{-1} . The additive in the electrolyte plays a vital role in suppressing

dendrite growth, thus improving the Zn stripping/plating CE. This is done by reducing the corrosion on zinc³² or enabling planar zinc deposition.⁵² For instance, according to Chen et al., the addition of Na_2SO_4 in ZnSO_4 electrolyte could effectively avoid the growth of Zn dendrites,⁵² because Na^+ with a lower reduction potential could form a positively charged electrostatic shield around the initial protuberances of Zn and avoid further dendrite deposition.¹⁴⁰ As a result, the case with Na_2SO_4 did not present a large number of vertical and harsh Zn dendrites formed on the surface of Zn anode (Figure 15E,F).

The use of aqueous electrolytes offers many advantages in terms of low cost, operational safety, facile manufacture, environmental benignity, and high ionic conductivities.

Very recently, Wang and colleagues developed a concentrated aqueous electrolyte consisting of both the lithium salt LiTFSI and the zinc salt ($\text{Zn}(\text{TFSI})_2$), where TFSI is bis(trifluoromethane) sulfonamide.³¹ Molecular dynamics (MD) studies indicate that in the most concentrated electrolyte (1 M $\text{Zn}(\text{TFSI})_2/20 \text{ M LiTFSI}$), the Zn^{2+} -solvation sheath is occupied primarily by TFSI, with six coordinating oxygens all from TFSI, rather than H_2O in the most dilute electrolyte (1 M $\text{Zn}(\text{TFSI})_2/5 \text{ M LiTFSI}$) (Figure 16A). This is very important to suppress the dendrite formation of Zn metal during cycling (Figure 16B), because strong interaction between Zn^{2+} and H_2O can prompt formation of inactive ZnO that limits reversibility by causing active material loss and dendrite growth on the Zn metal surface.³¹ This unique electrolyte with optimized Zn^{2+} solvation-sheath not only enables dendrite-free Zn plating/stripping at nearly 100% CE

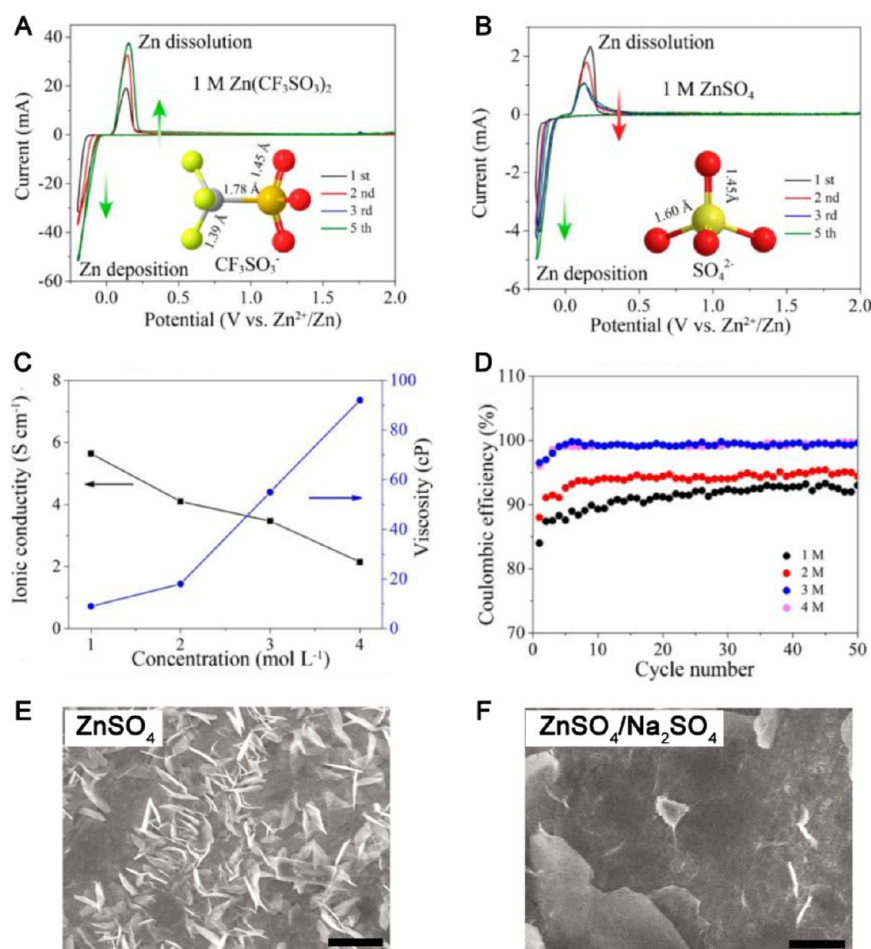


Figure 15. Cyclic voltammograms of Zn anode in (A) 1 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ and (B) 1 M ZnSO_4 at the scan rate of 0.5 mV s^{-1} . (C) Viscosity, ionic conductivity, and (D) Zn stripping/plating CE in aqueous $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ electrolyte with different concentrations (1–4 M). Reprinted from ref 33. Copyright 2016 American Chemical Society. SEM images (scale bars, $2 \mu\text{m}$) of Zn anode after one CV cycle in (E) ZnSO_4 and (F) $\text{ZnSO}_4/\text{Na}_2\text{SO}_4$ electrolytes. Reprinted with permission from ref 52. Copyright 2018 Macmillan Publishers.

(Figure 16C) but also retains water in the open atmosphere. This finding opens up a new strategy toward highly reversible Zn anode and accelerating the adoption of aqueous ZIBs for grid-scale electrochemical energy storage.

Summary and Perspectives. In this Review, we focus on the recent significant progress of aqueous ZIBs and provide in-depth discussions of main aspects involved in their development. Particularly, the main three energy storage mechanisms in aqueous ZIB systems are summarized: (1) Zn^{2+} insertion/extraction reaction mechanism, (2) chemical conversion reaction mechanism, and (3) $\text{H}^+/\text{Zn}^{2+}$ coinserion/extraction reaction mechanism. A comprehensive understanding of the battery materials is also proposed. These include the merits and issues related to Zn anode, optimized electrolyte, and various types of cathodes, as well as the optimization strategies aiming at high-performance aqueous ZIBs.

However, many challenges still exist in making great breakthroughs in high-performance aqueous ZIBs for practical applications. Here, the deep insights of potential future research directions will provide a guide for the high-efficiency energy storage as follows:

In-Depth Understanding of Energy Storage Mechanisms. To date, many works have focused on energy storage mechanisms for aqueous ZIB systems; however, the detailed information about the intrinsic electrochemical reaction mechanisms in this

system is still under debate. For one thing, many of the guest ions in this system, such as Zn^{2+} , H^+ , or other cations (e.g., Li^+ , Na^+ , Mn^{2+}) from the electrolyte additive, could react with cathode materials; their different insertion thermodynamics and kinetics make it hard to determine the specific reaction. For another thing, various cathode materials all lack available data in terms of theoretical number of guest ions and their sites stored in the host of these cathode materials because of the complexity of reaction and the negative effects from side reactions. In this regard, more powerful tools, such as advanced characterization technologies and precise calculations, are urgently needed for further investigation and prediction to accelerate the understanding of currently unclear reaction mechanism in aqueous ZIBs.

To date, many works have focused on energy storage mechanisms for aqueous ZIB systems; however, the detailed information about the intrinsic electrochemical reaction mechanisms in this system is still under debate.

Electrolyte Optimization. The optimization of the electrolyte has been regarded as one of the most effective ways to establish

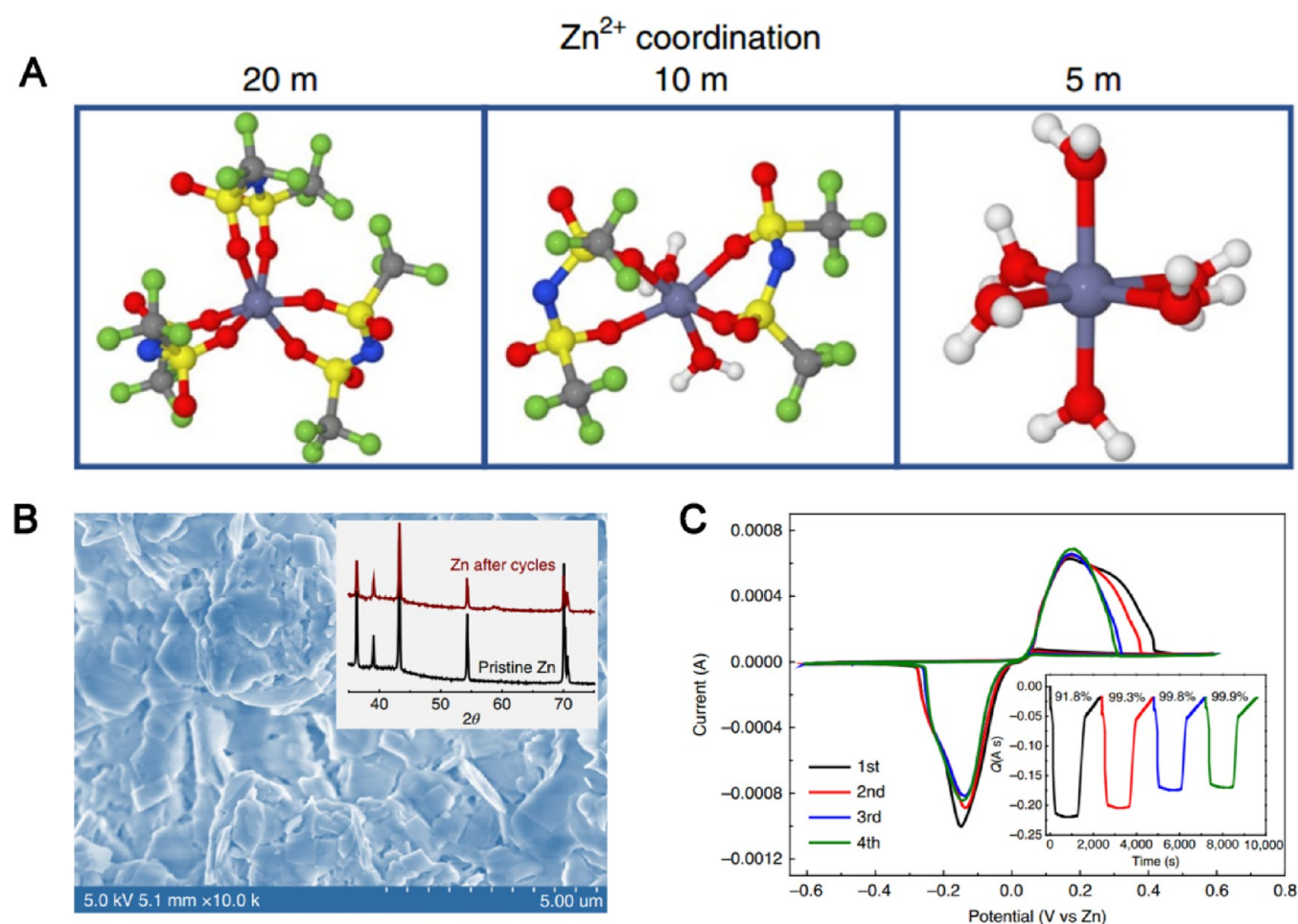


Figure 16. (A) Representative Zn²⁺-solvation structures in the electrolytes with 1 M Zn(TFSI)₂ and three concentrations of LiTFSI (5, 10, and 20 M). (B) SEM image and XRD pattern (inset) of a Zn anode after 500 stripping/plating cycles in 1 M Zn(TFSI)₂ + 20 M LiTFSI. (C) Cyclic voltammogram of Zn plating–stripping in a three-electrode cell at a scan rate of 1 mV s^{−1}. Inset: chronocoulometry curves. Reprinted with permission from ref 31. Copyright 2018 Macmillan Publishers.

a suitable environment that enables efficient Zn plating/stripping and stabilizes the cathode materials, which are dominated by the chemical and physical properties of the electrolytes such as solvents, zinc salts, salt concentrations, and additives. According to Wang and colleagues,³¹ the coordination of Zn²⁺ ions (or Zn²⁺-solvation sheath) in solution should be specially considered because of its effect on the dendrite formation of Zn metal during cycling. Therefore, the principles for development of aqueous electrolytes lie in adjusting their features that guarantee high Zn plating/stripping Coulombic efficiency and form an efficient Zn²⁺-solvation sheath to suppress dendrite growth on the Zn metal surface. In addition, a suitable and easy-to-handle electrolyte that is safe, nontoxic, eco-friendly, and inexpensive is urgently needed for grid-scale electrochemical energy storage.

Toward a Highly Reversible Zn Anode. The main issues such as formation of dendritic zinc, corrosion, and poor plating/stripping coulombic efficiency still plague the development of the Zn anode. As mentioned above, many research efforts focus on the electrolyte optimization and construction of novel Zn composite anodes; however, the discussion related to the design of novel monolithic Zn anode has rarely been reported. In nickel–zinc alkaline cells, Rolison et al. have designed a 3D monolithic zinc sponge anode that elevates the performance without undergoing passivation or macroscale dendrite

formation.²⁵ Therefore, to make improvements in aqueous ZIBs, these topics, such as construction of a three-dimensional Zn foam or sponge, should also be significantly addressed in future research. Also, motivated by suppressing Li dendrite growth via membranes or interfacial engineering,¹³⁰ the surface modification of Zn anode aiming to reduce the corrosion on zinc and support planar zinc deposition could be taken into account, but it should be based on the fast Zn plating/stripping.

Novel Strategies for Cathode Materials. It can be seen from the present Review that divalent Zn²⁺ ions have strong electrostatic interaction with many cathode materials in aqueous ZIBs, thus causing slow Zn²⁺ diffusion. These kinetic issues have also hindered other multivalent battery technologies, such as MIBs and AIBs,¹¹² which could be addressed by structural engineering and exploring new cathode materials. For structural engineering, because cation vacancy summarized in this Review shows the possibility of unlocking the electrochemical activity of well-known materials,³³ other structural defects (e.g., oxygen vacancy, ion doping, phase boundary, highly active facets, etc.) discussed in other battery systems^{141–145} could potentially provide new ideas for future generations of aqueous ZIBs. In addition, cathode materials with features of high porosity, large surface area, and enlarged interplanar crystal spacing may facilitate Zn²⁺ diffusion.^{146–148}

For exploring new cathode materials, according to the aforementioned discussion, many cathode materials, such as Mn-based oxides and V-based oxides, exhibit complicated electrochemical behaviors with phase transformation or formation of a Zn-containing phase during insertion/extraction of Zn^{2+} , which may be ascribed to the high electronegativity of oxygen.¹¹⁷ Therefore, the structures without oxygen (e.g., metal carbides, metal sulfides, and metal nitrides) can serve as an alternative to diminish this effect and are reactive enough to enable fast diffusion of Zn^{2+} , because these elements have electronegativity that is lower than that of oxygen.

It can be seen from the present Review that divalent Zn^{2+} ions have strong electrostatic interaction with many cathode materials in aqueous ZIBs, thus causing slow Zn^{2+} diffusion.

Designing Flexible Aqueous ZIBs. Flexible and wearable batteries may provide high flexibility and reliable electrochemical properties for portable and flexible electronics.¹⁴⁹ Recently, flexible aqueous ZIBs with gel or solid-state electrolyte have shown great potential in this field,^{52,56,133–135} because they may solve the intrinsic safety issues of LIBs with organic electrolytes. However, one of the main challenges is how to achieve flexible aqueous ZIBs with high zinc-ion conductivity, which quite depends on the two key components of the ZIBs (namely, electrolyte and electrode). For the electrolyte, the currently used polymer electrolytes (e.g., poly(vinyl alcohol) (PVA)-based electrolyte) mainly suffer from low ionic conductivity and poor mechanical strength. Therefore, new strategies are highly desired to achieve a well-designed flexible electrolyte with high zinc ion conductivity and superior mechanical properties for the further development of extremely safe and rechargeable ZIBs. For instance, Zhi et al. introduced a novel polyacrylamide (PAM)-based polymer electrolyte with an ultrahigh ionic conductivity of $1.76 \times 10^{-2} \text{ S cm}^{-1}$, remarkable flexibility, and favorable mechanical strength,¹³⁴ suggesting the selection of polymer host is a good choice. For electrodes, they are generally limited by the poor mass diffusion of divalent cation Zn^{2+} in the bulk lattice; therefore, more attention should be paid to improving the reaction kinetics of Zn^{2+} . In addition, the designed electrodes should be compatible enough with the polymer electrolyte to reduce the interface barrier of ionic transport.

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